Agreement-Stratified Evaluation of Deep Learning for Anatomical Landmark Recognition in Upper Gastrointestinal Endoscopy

A thesis submitted in partial fulfilment of the requirements
for the degree of B.Sc. in Computer Science and Engineering

Submitted by

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Corpus: GastroHUN — 8,834 images · 387 patients · 23 classes · 4 independent annotators

Approval

This thesis titled “Agreement-Stratified Evaluation of Deep Learning for Anatomical Landmark Recognition in Upper Gastrointestinal Endoscopy”, submitted by Fatin Sadab Nibirh (ID: 0242310005101526) and MD Himel Rahman (ID: 0242310005101800) to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfilment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents.

Table 1. Board of examiners.

Role	Name and affiliation	Signature
Supervisor	Ms. Shayla Sharmin Assistant Professor Department of Computer Science and Engineering Daffodil International University	
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Internal Examiner		
External Examiner		
Chairman		

Declaration

We, Fatin Sadab Nibirh and MD Himel Rahman, hereby declare that this thesis has been prepared by us under the supervision of Ms. Shayla Sharmin, Assistant Professor, Department of Computer Science and Engineering, Daffodil International University, and co-supervised by Dr. Md. Zahid Hasan, Associate Professor. We further declare that neither this thesis nor any part of it has been submitted elsewhere for the award of any degree or diploma.

The work reported here was carried out on the publicly released GastroHUN corpus [6], distributed under CC BY 4.0 with ethics approval CEI-2019-06-10 recorded by its custodians. No new patient data were collected and no identifiable data were handled at any point.

On the provenance of every number in this document

Every quantity reported in this thesis was resolved from a committed JSON artefact by a script, and every figure was drawn by a committed script. No number and no reference in this document was typed by hand. Appendix E lists the scripts and Appendix F the citation provenance, so any claim here can be traced to the code that produced it.

Table 2. Declaration signatories.

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Acknowledgements

We thank our supervisor, Ms. Shayla Sharmin, and our co-supervisor, Dr. Md. Zahid Hasan, for holding this project to a standard it would not otherwise have reached. The insistence that a claim be pre-registered before it is tested, and withdrawn when it does not survive, shaped the thesis more than any single result in it.

We thank the custodians of the GastroHUN corpus [6] for releasing the individual annotators' labels rather than a single consensus. That decision is what makes this thesis possible: without per-annotator labels, not one endpoint in this design exists. We thank the teams behind HyperKvasir [5] and GastroVision [38] for the same reason at the external-validation stage.

We thank the Department of Computer Science and Engineering at Daffodil International University for the compute on which every experiment reported here was run, and our families for their patience across the months this work took.

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List of Abbreviations

Table 3. Abbreviations and symbols used in this thesis.

Abbreviation	Meaning
AURC	Area Under the Risk–Coverage curve
CAM	Class Activation Map (Grad-CAM, gradient-weighted)
CI	Confidence Interval
CNN	Convolutional Neural Network
ECE	Expected Calibration Error
EGD	Esophagogastroduodenoscopy (upper GI endoscopy)
GI	Gastrointestinal
IoU	Intersection over Union
LOO	Leave-One-Out (annotator held out from the reference panel)
MCE	Maximum Calibration Error
OOP	Out-Of-Protocol (an image that is not an SSS station)
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RQ	Research Question
SSS	Systematic Screening of the Stomach protocol
α	Krippendorff's alpha, a chance-corrected agreement coefficient
AC1	Gwet's AC1, a chance-corrected agreement coefficient
κ	Cohen's or Fleiss' kappa, a chance-corrected agreement coefficient
ε	Label-smoothing strength
λ	Weight on the anatomy-aware penalty term

Abstract

Background. Deep classifiers for endoscopic landmark recognition are trained and evaluated almost exclusively on images where expert annotators agree. In this corpus that subset is 60.2% of the whole. What happens on the remainder is not reported, because the per-annotator labels needed to ask are rarely published.

Objective. To separate two quantities that the literature reports as one: the degradation of the classifier, and the degradation of the reference standard against which it is scored.

Methods. A ConvNeXt-Tiny baseline was reproduced, then evaluated across four strata of annotator agreement on the full 1,353-image test split. Five target constructions were trained on a cohort held constant, including a label-smoothing control whose strength was derived rather than chosen. All five arms were transferred to two external corpora without adaptation. Finally each annotator was held out in turn and scored against the other three by the identical metric applied to the model, alongside the modal-vote oracle that bounds any single-label predictor. Every phase was pre-registered before it ran; every internal interval is a patient-clustered bootstrap of 1,000 resamples.

Results. Annotator-marginalized macro F1 falls 83.9 → 48.9 → 26.2 across the first three strata and stands at 30.8 on the fourth, which is not lower than the third because its attainable ceiling is different and it holds only 81 images; the ceiling falls with the score throughout, and the ceiling-normalised unanimous-minus-majority gap is 17.98 points (95% CI [12.49, 24.37]). Expected calibration error rises 9.2% → 56.4%: mean confidence falls only 9.34 points while expected accuracy falls 56.57. No target construction repairs this: the pre-registered contrast against the matched control is 1.40 points ([-0.88, 3.68]). On contested images the model out-predicts a held-out annotator (0.4575 against 0.3902) but recovers only 24% of the headroom to the modal-vote oracle (0.6700) and exceeds that oracle on no stratum. Externally the arms are separated decisively by selective prediction (AURC 0.1906 for C2 against 0.3235 for C3) where every internal endpoint failed to separate them.

Conclusion. Agreement stratification separates a falling reference standard from a falling classifier. Both fall; the ceiling accounts for the larger share and a real model shortfall remains. What degrades further and faster than discrimination is confidence, and the endpoint that exposes it is the model's willingness to decline rather than its accuracy. A negative-control audit of the retired corpus (PARTIALLY SUPPORTED) shows the audit protocol that made these claims trustworthy is a well-formedness instrument, not a viability one.

1. Introduction

1.1 Clinical context

The Systematic Screening of the Stomach protocol prescribes a fixed set of photodocumented anatomical stations so that a gastroscopy can be shown to have been complete. Automating the recognition of those stations is a natural target for deep learning: the classes are defined, the images are routinely captured, and completeness is a measurable quality endpoint [15, 57, 79]. This thesis works with GastroHUN [6], 8,834 images from 387 patients, labelled independently by 4 annotators across 23 classes.

1.2 The problem: ground truth is a constructed object

Those 4 annotators agree unanimously on 60.2% of the corpus. Published baselines for this dataset [6] — and, as the review in Chapter 4 shows, for this literature generally — are trained and evaluated on that subset. The remaining 39.8% is not reported on, and 6.95% of images admit no majority label under any voting rule. A model's performance on the agreed subset is therefore not an estimate of its performance in a clinic, and the difference is not a matter of degree: on the hardest images there is no single label for the model to be right about.

The central question

The question this thesis exists to answer is not 'how accurate is the classifier'. It is: when performance falls as annotators disagree, how much of that fall is the classifier getting worse, and how much is the reference standard ceasing to exist? Those are different quantities with different consequences, and the literature reports their sum.

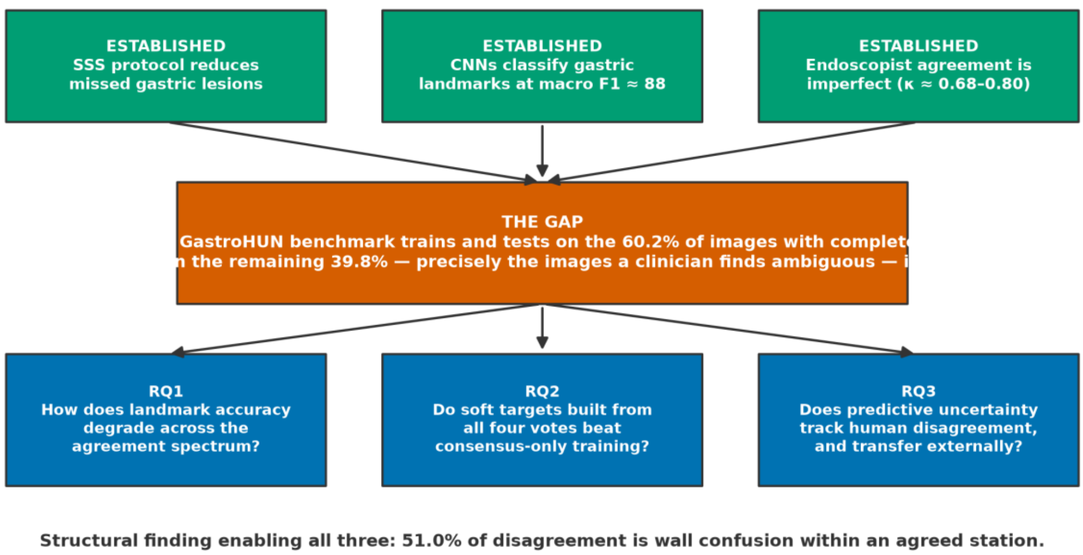


Figure 1. Ground truth as a constructed object: the same images admit different reference standards depending on how annotator votes are combined.

1.3 The central claim

Central claim

Expert-agreement stratification separates two things the literature reports as one: a reference standard that ceases to exist, and a classifier that falls short of what remains of it. As agreement falls the attainable ceiling drops from 1.00 to 0.67, so most of the apparent collapse is the ceiling moving, while the model still recovers only 24% of the distance from an individual annotator to that ceiling. Confidence degrades further and faster than discrimination, no *target construction repairs it, and the endpoint that separates configurations is not accuracy but the model's willingness to decline.*

1.4 Contributions

- An agreement-stratified evaluation protocol that reports the attainable ceiling alongside the score, so that a falling reference standard is not mistaken for a falling model (Chapter 5).
- A human comparator built from held-out annotators, scored by the identical metric on the identical images, together with the modal-vote oracle that bounds it (Chapter 8). To our knowledge this comparison has not previously been reported for this task.
- Evidence that no target construction among five — including a control whose smoothing strength was derived to match the soft target's displaced probability mass [33, 73] — repairs calibration on contested images (Chapter 6).
- An out-of-protocol rejection and selective-prediction analysis showing that configurations indistinguishable internally separate decisively under domain shift (Chapters 7 and 8).
- A negative-control evaluation of the audit protocol itself, which showed it to be a well-formedness instrument rather than a viability one, and three proposed extensions (Chapter 9).

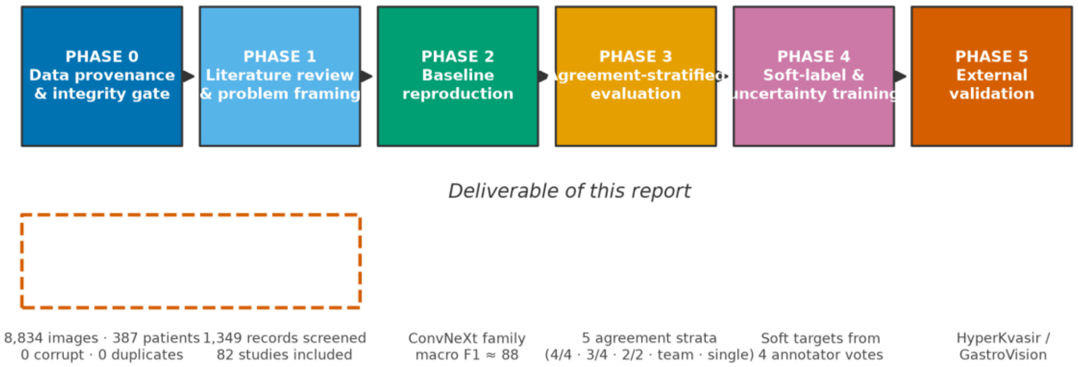


Figure 2. **Phase structure. Each phase gates the next; no phase's analysis was written before its pre-registration was frozen.**

2. Corpus Audit and Data Provenance

2.1 Why an audit chapter precedes any modelling chapter

Every claim in the chapters that follow is conditional on the corpus being what it says it is. The audit was therefore run first and deliberately measured no model performance, so that the hypotheses in Chapter 5 remained genuine predictions.

2.2 Integrity and provenance

Table 4. Corpus audit summary. Every value regenerates from src/data/gastrohun_*.py.

Measurement	Value
Manifest rows	8,834
Decoded successfully	8,834
Missing / orphan / corrupt	0 / 0 / 0
Patients	387
Classes	23
Annotators	FG1, FG2, G1, G2
Near-duplicate pairs examined	39,015,361
Cross-split duplicates, uncalibrated rule	54
Cross-split duplicates, calibrated rule	0
Split class-composition χ^2 p	1.00000
Classes below ±10 pp Wilson half-width	22/23
Patients with a clinical record	60.21%

A methodological lesson carried into every later threshold

The contamination scan first reported 54 cross-split duplicate pairs. That number was an artefact of an uncalibrated threshold. A first correction using a randomly-paired null was also wrong, because random pairs mostly compare different anatomical stations and set an artificially low bar. Rebuilding the null from class-matched pairs and anchoring the rule on a synthetic-duplicate positive control reduced the count to 0, and visual audit confirmed the flagged pairs were different patients photographed at the same landmark. An uncalibrated threshold is not a measurement, and a threshold calibrated against the wrong comparison is not much better.

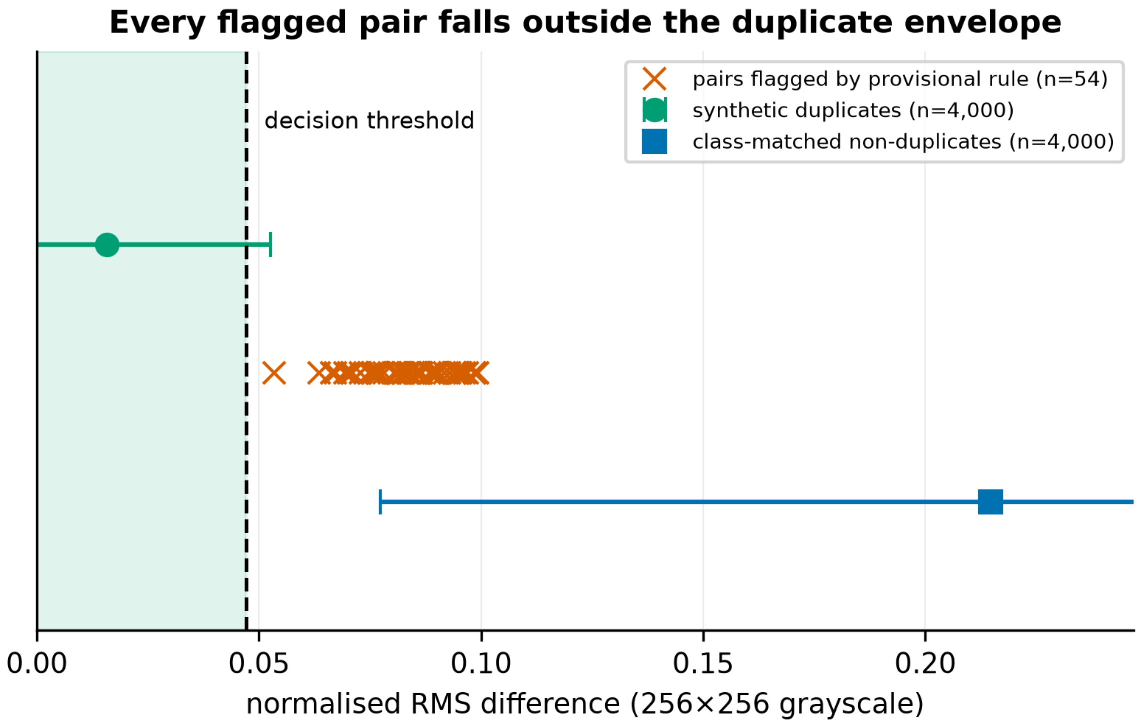


Figure 3. Threshold calibration. The positive control fixes where the decision rule belongs; the class-matched null shows why the first attempt was wrong.

2.3 Agreement, and why three chance corrections coincide

Fleiss' κ = 0.7475 [19], Krippendorff's α = 0.7475 [30] and Gwet's AC1 = 0.7475 [29]. The three coincide because the SSS protocol makes the class marginal near-uniform, so $\Sigma p_j^2 \approx 1/K$ and the Fleiss and Gwet expectations become algebraically equal. The kappa paradox does not apply here.

Table 5. Pairwise agreement. Each resident agrees more with the gastroenterologists than with the other resident, so seniority does not predict agreement and no design may treat a team as a coherent unit.

Annotator pair	Cohen's κ [10]	95% CI	Relationship
G1-G2	0.8031	[0.7890, 0.8171]	within team
FG1-G1	0.7930	[0.7768, 0.8082]	between teams
FG1-G2	0.7627	[0.7453, 0.7804]	between teams
FG2-G1	0.7260	[0.7095, 0.7432]	between teams
FG2-G2	0.7209	[0.7069, 0.7336]	between teams
FG1-FG2	0.6799	[0.6618, 0.6989]	within team

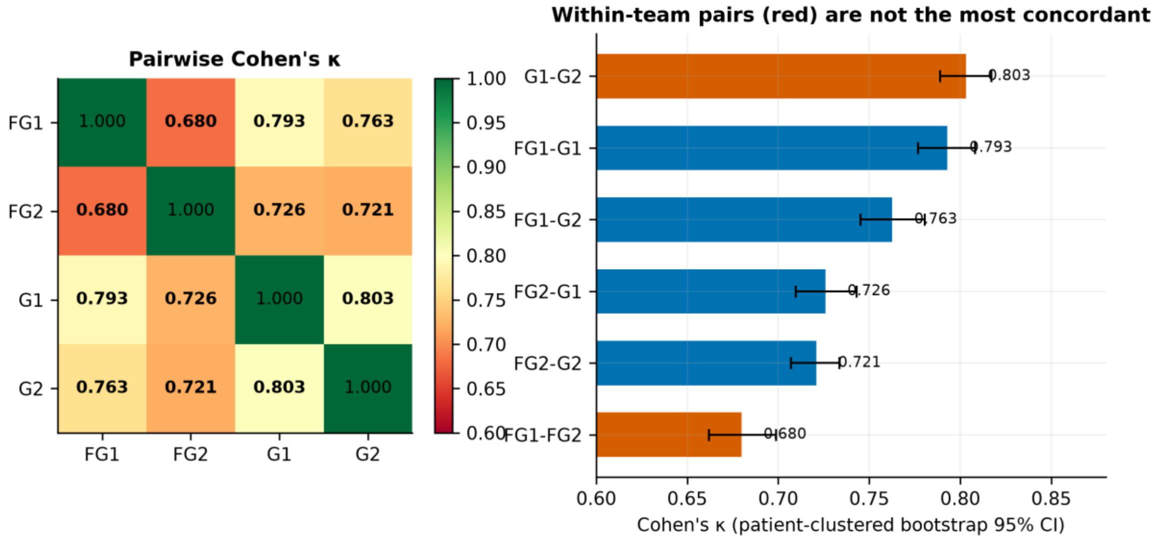


Figure 4. Pairwise agreement structure.

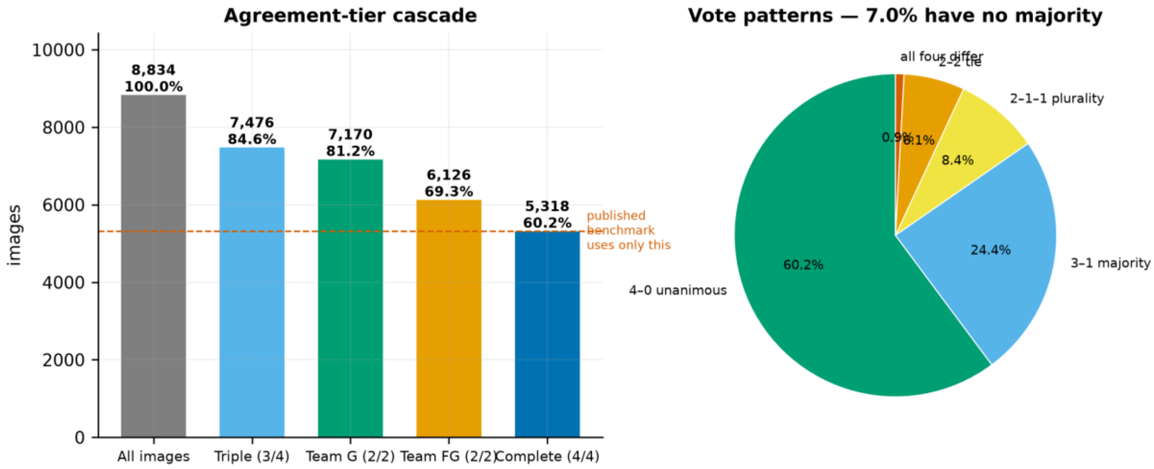


Figure 5. The agreement cascade. Published baselines for this corpus are computed on the unanimous tier alone.

2.4 Disagreement is anatomically structured

Table 6. Disagreement decomposition. Collapsing the wall axis recovers little; collapsing the station axis recovers a great deal.

Disagreement type	% of 12,800 events
same station different wall	50.96%
landmark vs OTHERCLASS	20.80%
same wall different station	19.91%
different wall and station	8.33%

Table 7. Agreement recomputed at coarser granularity. Endoscopists know how deep the scope is and disagree about which way it points — which is what makes the disagreement modellable rather than noise.

Granularity	Mean pairwise κ	Categories	Unanimity
Full	0.7476	23	60.20%
Station	0.8597	7	78.99%
Wall	0.7481	5	66.86%

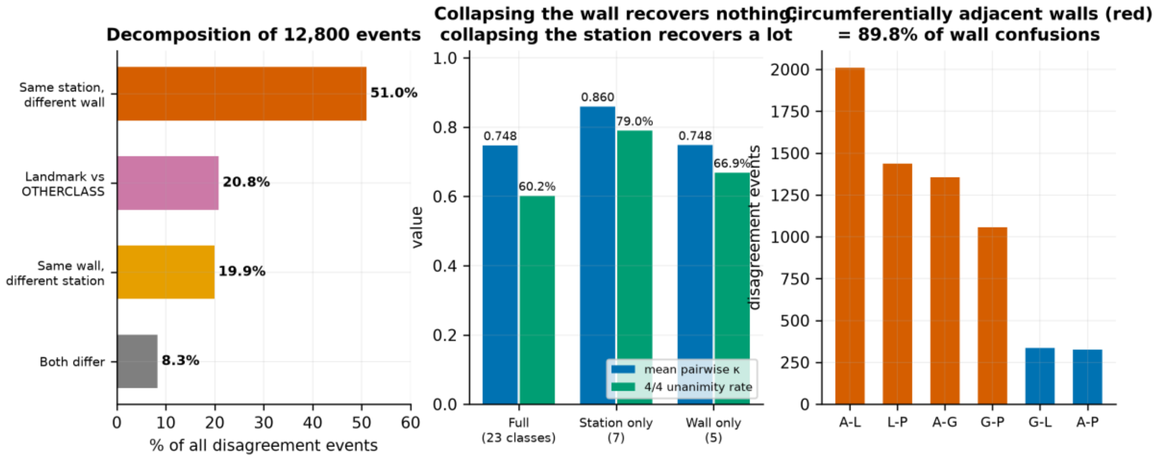


Figure 6. Where the disagreement lives.

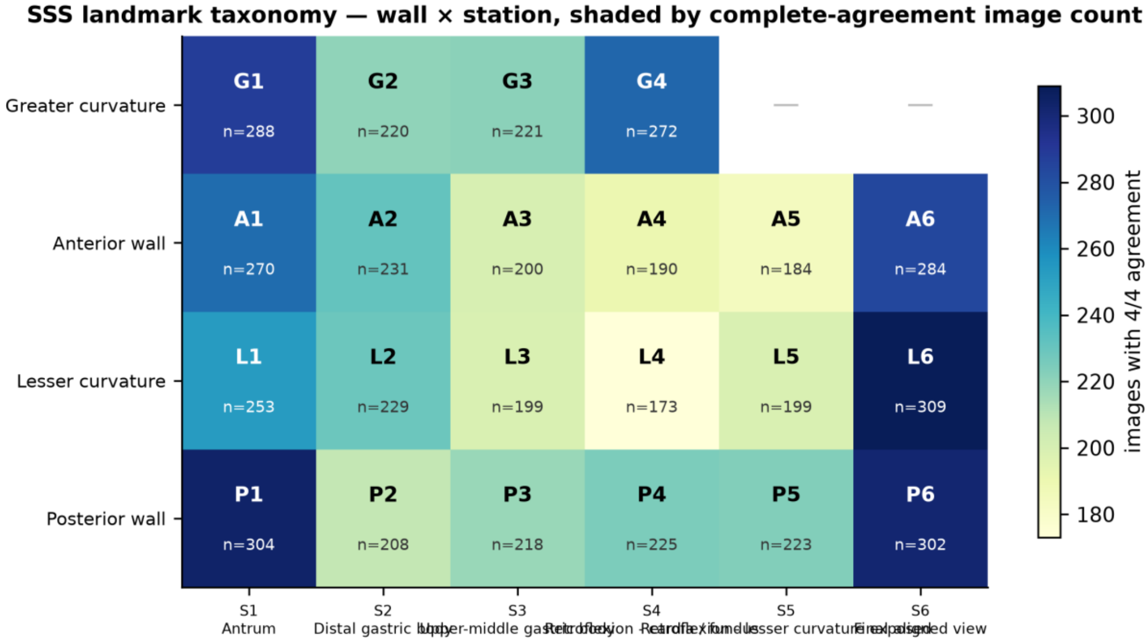


Figure 7. The wall × station grid. This structure is the thesis's main analytical lever and is not documented as such in the dataset descriptor.

2.5 OTHERCLASS is a subjective judgement

Per-annotator rejection rates span 1.48% to 8.90%, a 6.0× spread. Quality assessment and anatomical classification are different tasks and are modelled and evaluated separately throughout.

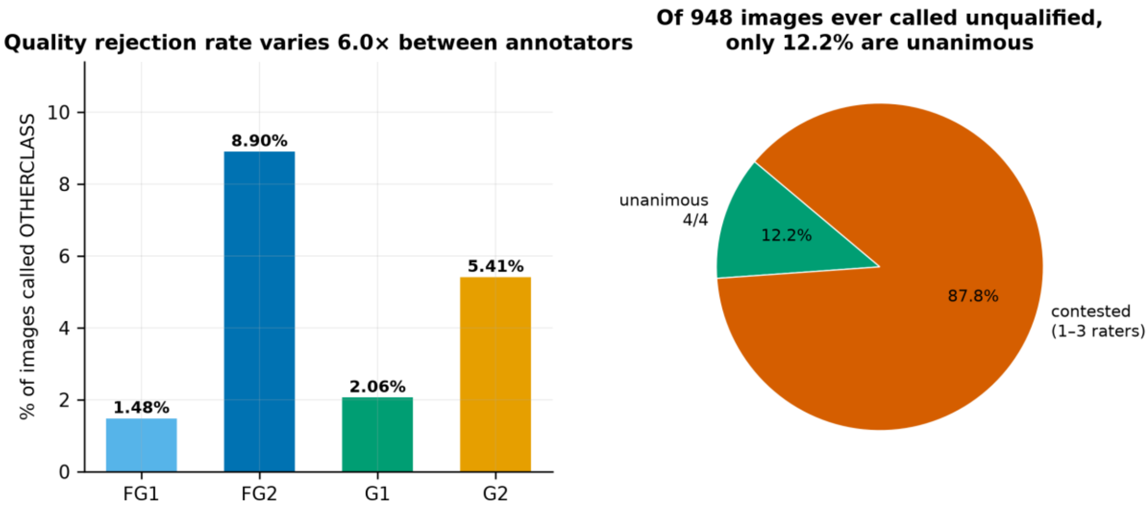


Figure 8. Per-annotator rejection behaviour.

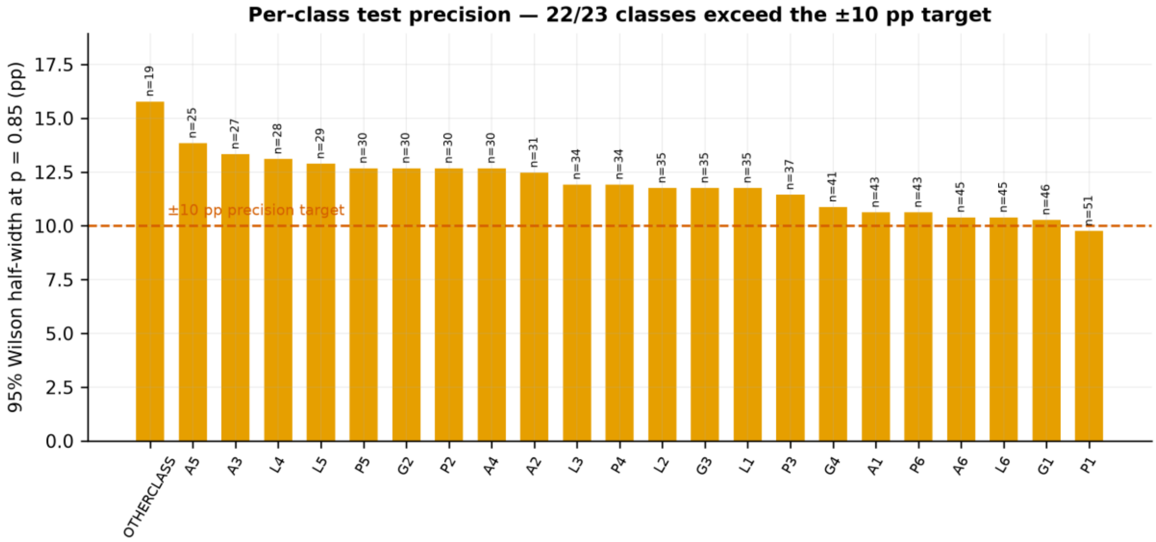


Figure 9. Per-class test-set precision. 22 of 23 classes exceed a ±10 pp Wilson half-width, which is why every per-class result in this thesis is exploratory.

3. Materials and Methods

This chapter fixes the machinery. Everything in it was frozen before the first result in Chapter 5 was computed, and every value below is read from the run artefact rather than transcribed into prose, so the description cannot drift from the experiment it describes.

3.1 Cohort and splits

The corpus holds 8,834 images from 387 patients [6]. The reproduction cohort is the complete-agreement subset — the 60.2% of images on which all four annotators agree — because that is the condition under which the published baseline was trained and scored, and Chapter 5 cannot claim to have reproduced a result it did not reproduce the conditions of. It comprises 3,722 training, 793 validation and 803 test images. The stratified evaluation in Chapter 5 then scores the frozen model on the full 1,353-image test split, contested images included.

Table 8. Split construction. Splits are patient-disjoint by construction, and class composition does not differ across them. Acquisition stream does differ, which is declared here rather than discovered later: it is the reason Chapter 5 carries an acquisition-stream confound control.

Property	Value
Split unit	patient — no patient contributes to two splits
Train / validation patient overlap	0 patients
Train / test patient overlap	0 patients
Validation / test patient overlap	0 patients
Class composition across splits	$\chi^2 = 13.48$, $p = 1.00000$
Acquisition stream across splits	$\chi^2 = 72.38$, $p = 1.92e-16$
Cohort images resolved to a file on disk	5,318 of 5,318

3.2 Pre-processing

Images are resampled once to 224×224 with Lanczos resampling and cached, so that every arm in every later phase reads pixel-identical inputs. Normalisation uses channel statistics computed on the 3,722 training images of this cohort — mean (0.601, 0.369, 0.279), sd (0.305, 0.234, 0.191) — rather than the ImageNet constants, which differ from them by up to 0.127 in the mean. Endoscopic illumination is red-dominant and the ImageNet statistics do not describe it.

Why the standard augmentation recipe is not used

The augmentation recipe omits horizontal and vertical flips and any large rotation, which is a departure from the standard image-classification recipe and is deliberate. The label encodes a gastric WALL — anterior, posterior, lesser curvature, greater curvature — so a horizontal flip maps an image of one wall onto the appearance of another and silently relabels it. An augmentation that destroys the target is not regularisation. What remains is a mild scale and aspect jitter (RandomResizedCrop, scale 0.85–1.00, ratio 0.90–1.11) and a photometric jitter (brightness, contrast and saturation ±0.20, hue ±0.02), neither of which moves the anatomy.

3.3 Architecture and training

Table 9. Training configuration, read from reports/phase2_run_seed1.json. The same configuration is reused unchanged by every later arm, so that the only thing varying across the Chapter 6 configurations is the target.

Setting	Value
Backbone	ConvNeXt-Tiny [51], ImageNet-pretrained
Fine-tuning depth	3 of 8 feature modules unfrozen (94.5% of feature parameters, 26,287,872 of 27,818,592)
Optimiser	AdamW

Setting	Value
Learning rate, head	0.001
Learning rate, fine-tuning	0.0001
Weight decay	0.05
Batch size	24 (gradient accumulation ×1, effective 24)
Precision / memory format	float32, channels last
Warm-up epochs	10
Maximum fine-tuning epochs	51
Early-stopping patience	10 epochs on validation macro F1
Seeds	1, 2, 3
Model selection	validation macro F1 only; the test split is scored once, after selection

All runs used a single NVIDIA GeForce GTX 1650 with 4096 MiB of memory, on Windows-11-10.0.26200-SP0 with Python 3.14.5, torch 2.12.0+cu126 and torchvision 0.27.0+cu126. Peak observed VRAM was 2324 MiB. The 4 GB memory limit is the binding constraint behind the epoch cap declared in Chapter 6, and is recorded here so that the cap reads as a resource limitation rather than a design choice.

3.4 Metrics, and exactly how each is computed

Four quantities carry the thesis, and three of them are reported in the literature under names that hide a choice. The choices are made explicit here.

3.4.1 Annotator-marginalized macro F1

The primary discrimination metric is the mean, over the 4 annotators, of the macro F1 computed against that annotator's labels taken alone. Macro F1 is averaged over all 23 classes with zero division scored as zero, so a class absent from both the reference and the prediction contributes 0 rather than being dropped. Marginalizing over annotators rather than scoring against a consensus is what makes the metric defined on contested images at all: where the four annotators disagree there is no consensus label to score against, but each annotator's own labelling remains a valid reference.

3.4.2 Agreement strata

Table 10. The four agreement strata. They partition the test split and are defined by the vote pattern alone, so membership is fixed before any model is trained and cannot be influenced by a result.

Stratum	Vote pattern	n (test)	Definition
S-unanimous	4-0	803	all four annotators give the same label
S-majority	3-1	342	three agree, one dissents
S-plurality	2-1-1	127	two agree, two dissent separately
S-no-majority	2-2 or 1-1-1-1	81	no label holds a majority under any voting rule

3.4.3 Calibration

Expected calibration error is computed with 10 equal-width bins over the [0, 1] confidence range, as the support-weighted mean absolute difference between mean confidence and expected accuracy within each bin [27]. The bin count is stated because ECE is not comparable across studies without it: the same predictions binned more finely generally yield a larger ECE. Expected accuracy on a contested image is the fraction of the four annotators whose label the prediction matches, so the calibration target degrades gracefully rather than becoming undefined where consensus fails.

3.4.4 Two attainable ceilings, and why they are not one quantity

A score means nothing on a contested image without knowing what the best possible score there would be. Two such bounds appear in this thesis. They were previously both called “the modal-vote oracle”, they differ by up to 14 points, and separating them is necessary before either can be read.

Table 11. The two bounds, recomputed from the committed votes by src/models/phase8_oracle_reconcile.py, which reproduces both committed series (9 of 9 checks). The panel ceiling is the modal label of all four votes scored against all four annotators, and bounds Chapter 5, where nothing is held out. The leave-one-out oracle is the modal label of three votes scored against those same three, and bounds Chapter 8, where it must face the identical task as the held-out human.

Stratum	Panel ceiling	LOO oracle	Difference	Mean tie depth, 4→3 refs
unanimous	1.0000	1.0000	+0.0000	1.00 → 1.00
majority	0.7423	0.7423	+0.0000	1.00 → 1.00
plurality	0.4456	0.4492	+0.0037	1.00 → 2.00
no-majority	0.4023	0.5452	+0.1429	2.20 → 1.20
contested (pooled)	0.6476	0.6697	+0.0221	1.18 → 1.26

Why the two bounds differ, measured rather than argued

The two coincide on S-unanimous and S-majority to five decimal places, and diverge by 14.29 points on S-no-majority. The cause is tie multiplicity, and it is a property of the vote pattern rather than of any model. S-no-majority is 73 images split 2-2 and 8 split 1-1-1-1; on a 2-2 image the modal label of four references is a two-way tie worth 2/4, and against any three references it becomes a 2-1 split with a unique mode worth 2/3, so removing a reference makes the oracle's task strictly easier. On S-plurality the effect cancels exactly — dropping one of the two modal voters leaves a three-way tie worth 1/3, dropping either singleton leaves a unique mode worth 2/3, and the four folds average to 0.5000 either way — so the residual 0.37 points there is not a difficulty difference at all but the residue of macro F1 being a nonlinear class-wise aggregate rather than expected accuracy. RQ1's confirmatory endpoint uses only the two strata where the definitions agree, so it is unaffected by the choice.

3.4.5 Intervals

Every internal interval in this thesis is a patient-clustered bootstrap of 1,000 resamples: the patients in the relevant split are resampled with replacement and the statistic is recomputed on each draw. Images are never resampled independently, because images from one procedure are not independent observations and an image-level interval on this corpus would be optimistically narrow. Where a ratio is reported, both numerator and denominator are recomputed inside each draw so that the pairing is preserved — in particular the ceiling-normalised gap in Chapter 5 resamples its ceiling rather than holding it fixed. The one place this construction is unavailable is external validation, where neither corpus publishes a case identifier; those intervals are image-level and are labelled as such wherever they appear.

3.5 Pre-registration and the gating discipline

Each phase writes a pre-registration file before it runs, fixing its hypotheses, its endpoints, its verdict rules and its analysis order. The generating scripts refuse to overwrite an existing pre-registration, so a rule cannot be revised after a result is seen; Appendix A lists the files and Appendix B every declared deviation. The Phase 2 rule is representative: “PASS if |observed seed-mean macro F1 - 85.0| <= 1.5 points; otherwise FAIL and the diagnostic order below is executed.” This is the discipline that lets Chapter 10 report three unresolved research questions as findings rather than as failures — each was a pre-registered test against a matched control, not a search that happened to come up empty.

4. Literature Review

A PRISMA 2020 review [60] across seven themed searches identified 1,382 records, 1,349 unique, of which 82 were included (68 from database searching, 14 hand-searched). 84.1% were published in 2020 or later, so the omissions catalogued below are properties of the current literature and not of its history.

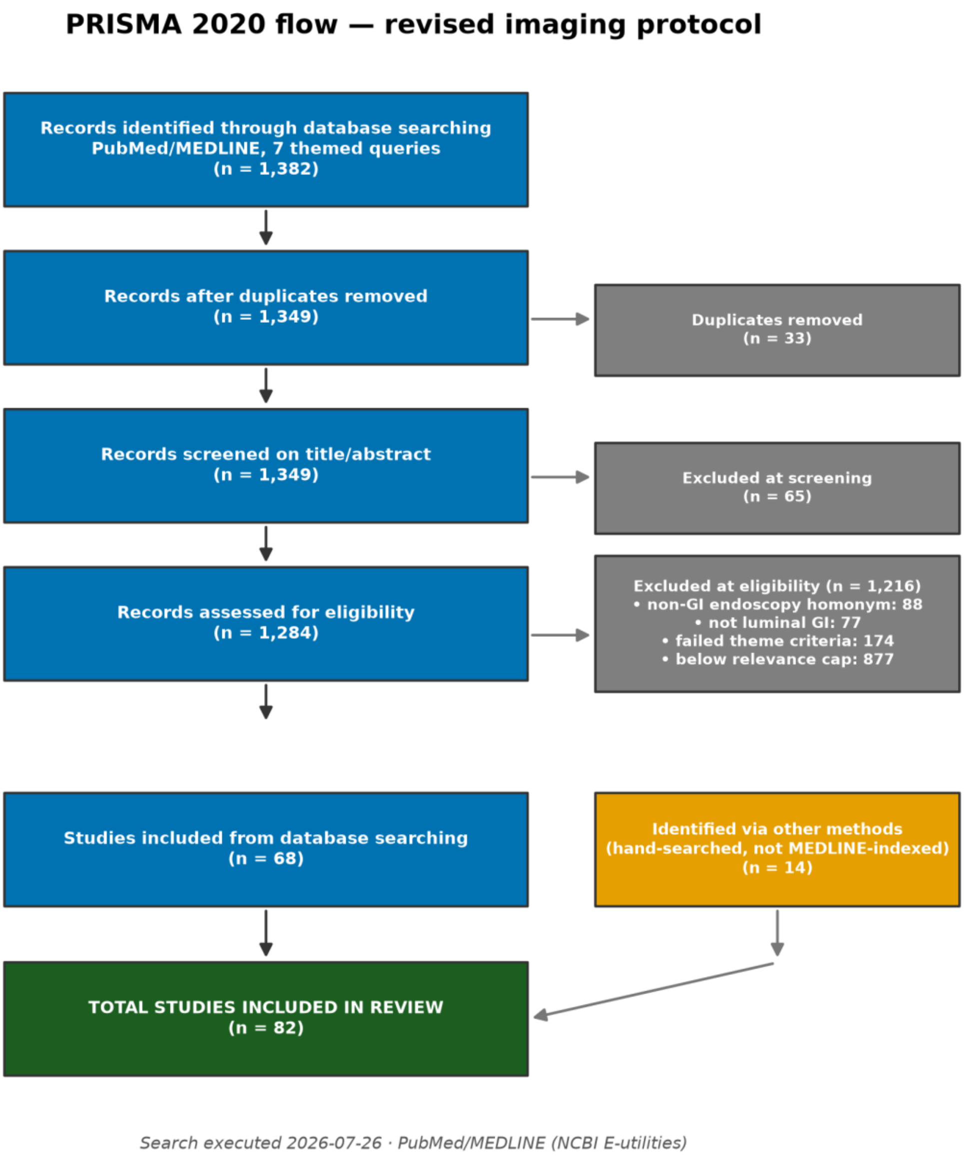


Figure 10. PRISMA 2020 flow.

4.1 What the seven themed searches cover

Table 12. The seven themed searches. Each addresses one link in the argument, and the review was structured this way so that a gap in the literature could be located to a specific link rather than asserted of the field as a whole. The final column lists every study the search contributed, so the reference list contains no entry that is not cited somewhere in the text.

	Themed search	n	Median yr	What it establishes	Included studies
T1	Landmark recognition in UGI endoscopy	17	2023	the task itself. Landmark and anatomical-site recognition in endoscopy, including the corpora this thesis transfers to.	[1, 5, 16, 24, 26, 31, 38, 40, 45, 48, 51, 53, 55, 59, 72, 78, 85]
T2	Endoscopic quality & blind-spot audit	10	2022	why the task matters. Blind-spot audit, photodocumentation completeness and quality control during endoscopy.	[9, 15, 47, 57, 62, 71, 79, 80, 81, 84]

	Themed search	n	Median yr	What it establishes	Included studies
T3	Inter-observer variability in endoscopy	14	2024	how much experts disagree, measured. Interobserver studies across endoscopic classification tasks.	[8, 10, 11, 14, 17, 19, 21, 28, 29, 30, 35, 43, 56, 63]
T4	Noisy / soft / multi-annotator labels	14	2023	what to do about disagreement. Noisy, soft and multi-annotator label learning.	[7, 12, 13, 22, 33, 37, 39, 46, 50, 66, 73, 75, 82, 86]
T5	Uncertainty & calibration	11	2024	whether a confidence can be believed. Calibration and predictive uncertainty.	[18, 20, 27, 36, 41, 42, 61, 65, 68, 76, 77]
T6	External validation & dataset shift	8	2025	whether any of it survives a second centre. External validation and dataset shift.	[2, 3, 4, 25, 49, 52, 64, 67]
T7	Reporting standards for medical AI	8	2025	what a study of this kind is obliged to report.	[23, 32, 34, 44, 54, 58, 70, 83]

4.2 What the included studies report, counted

The claim that this literature under-reports certain things is the claim the whole thesis rests on, so it is counted rather than asserted. Each of the 68 included studies that ships an abstract was screened by regular expression for five reporting dimensions. The counts are MENTION counts and are therefore upper bounds: a study that mentions calibration may not report it, but a study that never mentions it almost certainly did not make it an endpoint. The patterns were written generously, which weakens the bound rather than strengthening it.

Table 13. Reporting dimensions across the 68 included studies with abstracts. Generated by src/report/literature_synthesis.py.

Reporting dimension	Mentioning	%	Where this thesis addresses it
Population description (age / sex)	0/68	0.0%	NOT ADDRESSED -- the corpus ships no age or sex
Calibration of predicted probability	7/68	10.3%	Chapters 5 and 6 -- ECE by stratum as a primary endpoint
Predictive uncertainty	18/68	26.5%	Chapters 6 and 8 -- soft targets and selective prediction
External validation or shift	29/68	42.6%	Chapter 7 -- two external corpora, no adaptation
How the reference standard was built	29/68	42.6%	Chapters 3 and 5 -- the whole design

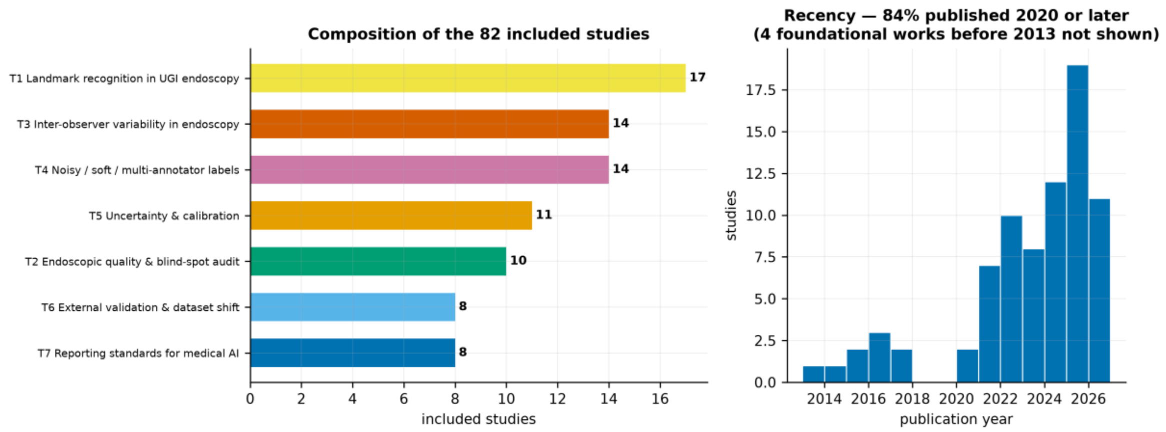


Figure 11. Reporting gaps in the included studies, and which this design addresses.

What these counts can and cannot bear

Two limitations travel with these counts and neither is cosmetic. 14 of 82 included studies ship no abstract in the MEDLINE record and are excluded from the denominator rather than counted as absences. And an abstract-level screen is not a full-text appraisal — which matters most for population description, where the count of 0/68 should be read as 'demographics are not a headline in this literature' and not as 'no study describes its population'. Cohort demographics live in a baseline-characteristics table, not in an abstract. The calibration figure carries no such caveat: a study reporting calibration as an endpoint names it in the abstract, so 10.3% is a firm ceiling on how much of this literature reports whether its probabilities can be believed.

4.3 The gap this thesis occupies

The four commonest omissions are missing external validation, absent calibration reporting, unexamined ground-truth construction, and incomplete population description. This design addresses three directly. The fourth cannot be met on this corpus — GastroHUN ships no age or sex — and is declared as a limitation in Chapter 10 rather than passed over.

The gap is sharper than any single dimension, and it is an intersection. Of the 29 studies that engage at all with how their reference standard was built, 5 — 17.2% — also say anything about whether their predicted probabilities are trustworthy [13, 22, 66]. The two questions are treated as unrelated. This thesis argues they are the same question: a confidence is a claim about a reference standard, and where the reference standard is contested the claim has no fixed referent.

> **The gap**
> *Ground-truth construction is the gap this thesis occupies. Of the included studies, those that used multiple annotators almost universally reduced them to a consensus label before modelling and reported no analysis of the images where consensus failed [8, 11, 14, 28]. The per-annotator labels required to do otherwise are rarely published; GastroHUN [6] is unusual in retaining them, which is why this design is possible on this corpus and on very few others.*

4.4 What is already known, and what this thesis does not claim

Two of this thesis's findings have antecedents, and saying so is what fixes the size of the contribution. That modern networks are overconfident, and that their calibration degrades under distribution shift, is established [20, 27, 42]. That training on soft or multi-annotator targets can transfer information a hard consensus label destroys is likewise established [13, 22, 33, 73]. Neither phenomenon is claimed here as new.

What is new is the measurement. No study in this review reports performance or calibration as a function of how much the expert panel agreed, because doing so requires per-annotator labels that are almost never released; and none separates the degradation of the classifier from the degradation of the reference standard it is scored against. That separation — the attainable ceiling reported beside the score — is the contribution, and it is a methodological one rather than a performance one. This thesis produces no model that classifies better than the published baseline, and does not claim to.

5. Agreement-Stratified Evaluation

5.1 Baseline reproduction as a validity check

Before anything was changed, the published ConvNeXt-Tiny [51] result was reproduced on the unanimous subset. Three seeds gave a mean macro F1 of 83.92 against a published ~85.0, inside the pre-registered ±1.5-point band. The pipeline was therefore validated before it was used to make any novel claim.

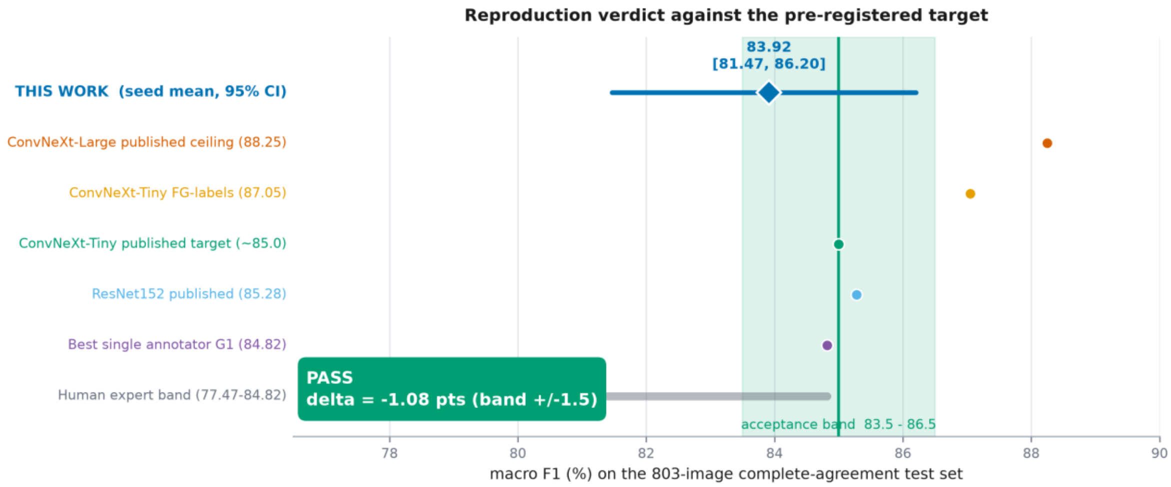


Figure 12. Baseline reproduction against the published target.

5.2 Results: the raw decline, and why it misleads

Table 14. Per-stratum performance with the attainable ceiling. The ceiling is the modal-vote oracle: the best any single-label predictor can achieve against the annotator panel on those images.

Stratum	n	Macro F1	Attainable ceiling	ECE %
Unanimous (4/4)	803	83.9	1.0000	9.2
Majority (3/4)	342	48.9	0.7423	38.2
Plurality (2-1-1)	127	26.2	0.4456	56.4
No majority	81	30.8	0.4023	49.1

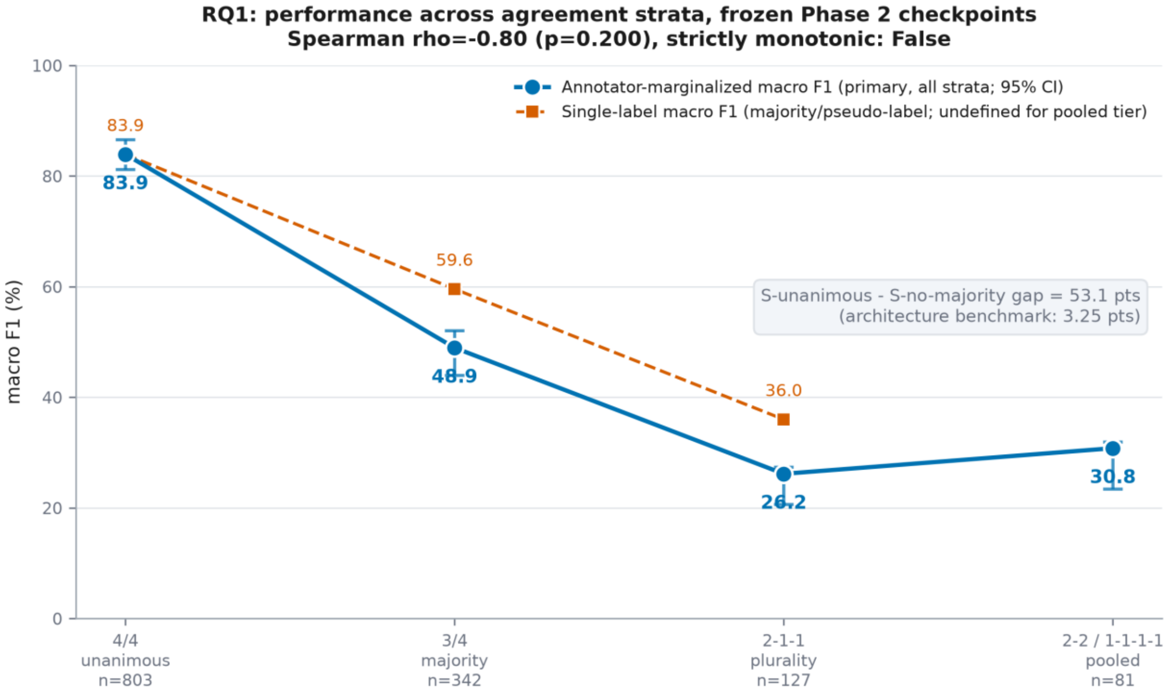


Figure 13. The raw stratified decline — the quantity the literature would report.

> **Ceiling normalisation, and why it is not optional**
>
> *Most of that decline is the ceiling moving. Held against the attainable maximum, the unanimous-minus-majority gap is 17.98 points (95% CI [12.49, 24.37]) rather than the 34.97 points the raw scores suggest. Reporting the raw decline alone would attribute to the model a change that is mostly a property of the reference standard.*

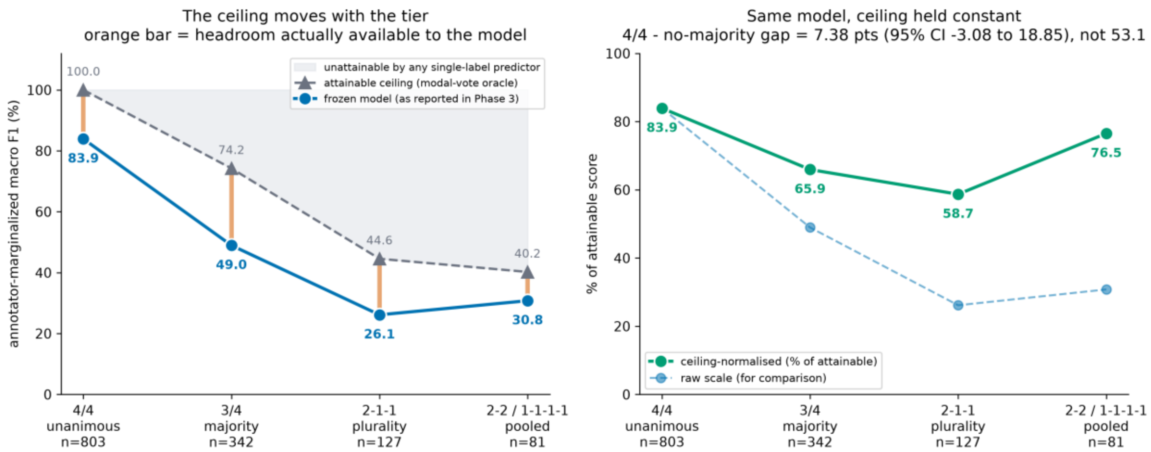


Figure 14. The same strata with the attainable ceiling held constant.

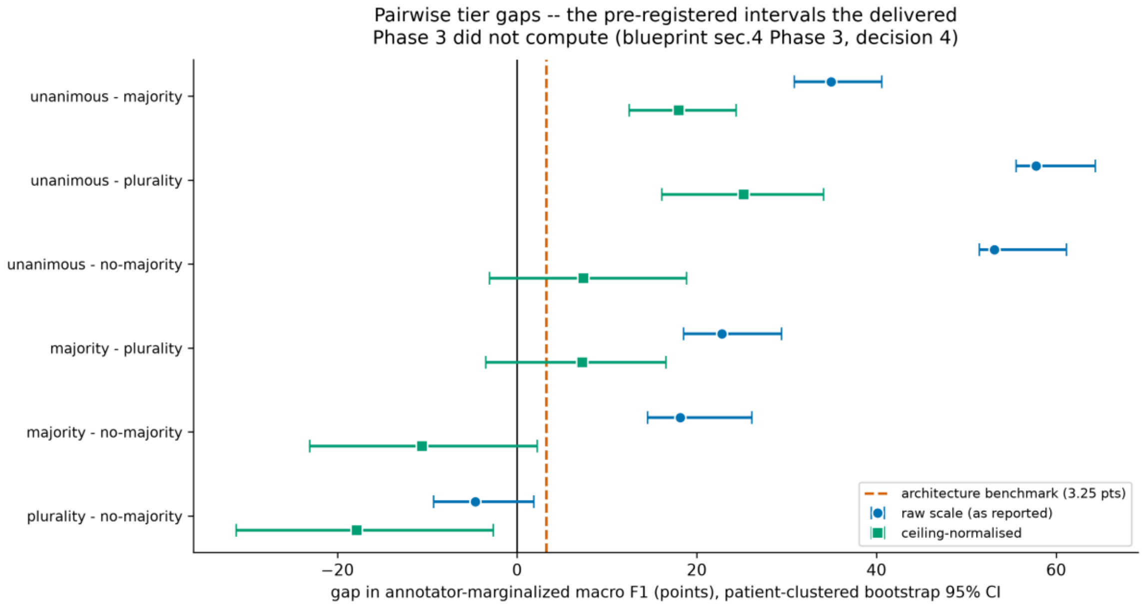


Figure 15. RQ1's confirmatory endpoint with patient-clustered 95% intervals.

5.3 The calibration collapse

Expected calibration error rises from 9.2% on unanimous images to 56.4% on the 2-1-1 stratum. The mechanism is visible in the components: mean confidence falls 9.34 points across those strata while expected accuracy falls 56.57. The model becomes wrong far faster than it becomes hesitant. This is the durable finding of the thesis and it survives every subsequent attempt to remove it.

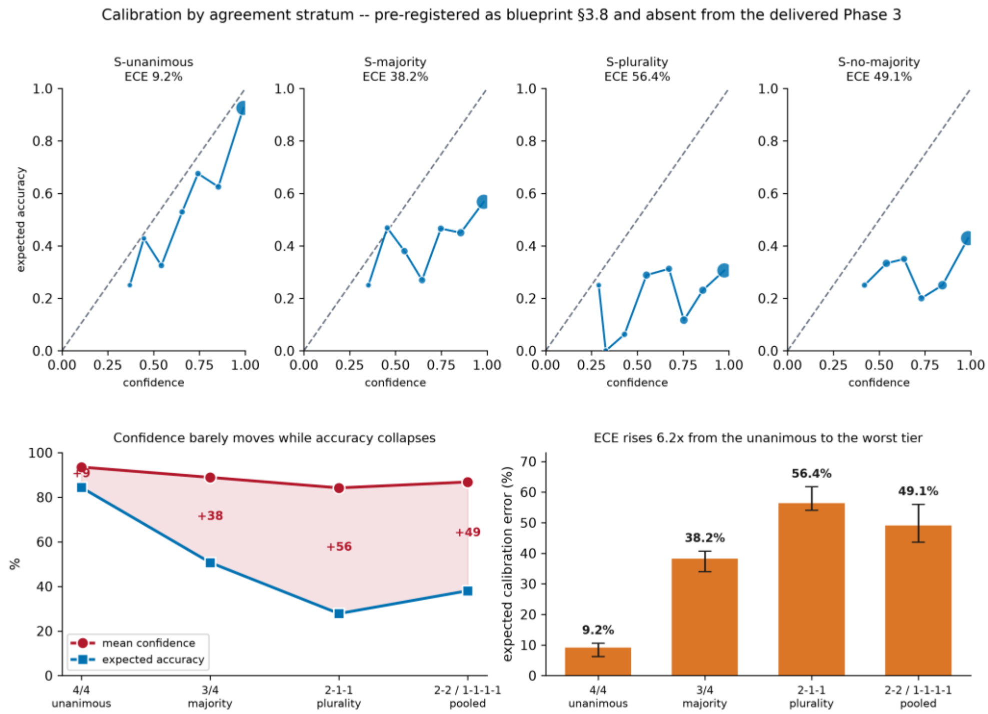


Figure 16. Calibration by agreement stratum.

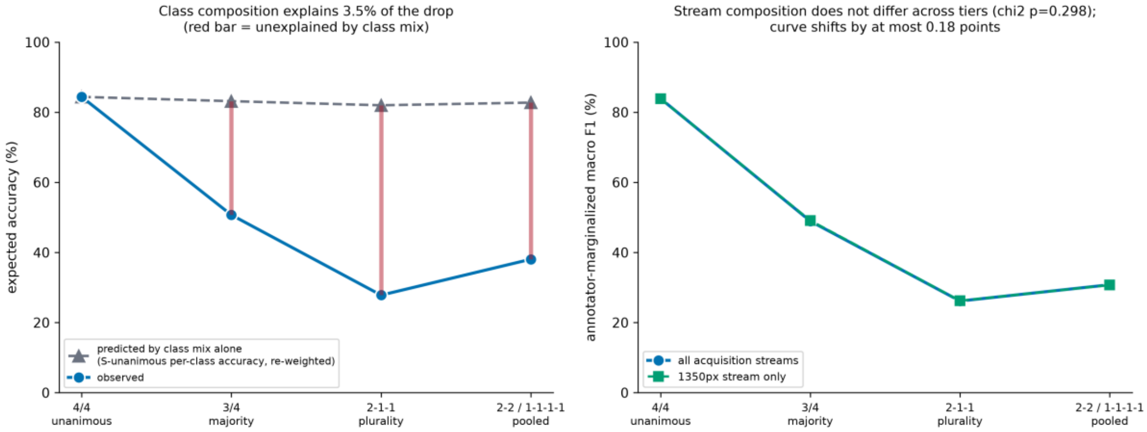


Figure 17. Confound controls. Class composition explains 3.5–4.2% of the drop and acquisition-stream composition does not differ across strata.

5.4 Corrections to the first analysis

Four claims made in the first version of this analysis did not survive re-examination and were withdrawn. They are recorded here rather than in an appendix, because a reader meeting the original claim elsewhere in the literature should meet the correction in the same place.

- The assertion that the any-annotator hit rate 'shows no such reversal' was false; the same dip is present in the published table.
- Any-annotator hit rate is confounded by tier-varying acceptance-set size and cannot be read as a skill measure across tiers.
- The confusion-geometry comparison against the human benchmark was withdrawn to hypothesis status because a model interval was compared against a human point estimate. Chapter 8 settles it.
- The '16× the architecture benchmark' headline was a scale artefact. The direction survives; the magnitude does not.

6. Target Construction and Uncertainty

6.1 Five configurations, one cohort, one thing varying

Table 15. The configuration matrix on the pooled contested stratum. Backbone, schedule, augmentation, normalisation and selection criterion are identical across arms; only the target differs.

Arm	Target construction	Macro F1 (contested)	ECE % (contested)	ECE % (unanimous)
C0	C0 hard label, 4/4 cohort	42.4	44.0	9.2
C1	C1 hard majority label	43.8	43.6	9.5
C2	C2 vote proportions	45.8	33.1	4.7
C3	C3 hard + matched smoothing (control)	44.4	19.4	6.5
C4	C4 vote proportions + anatomical penalty	45.5	32.7	4.1

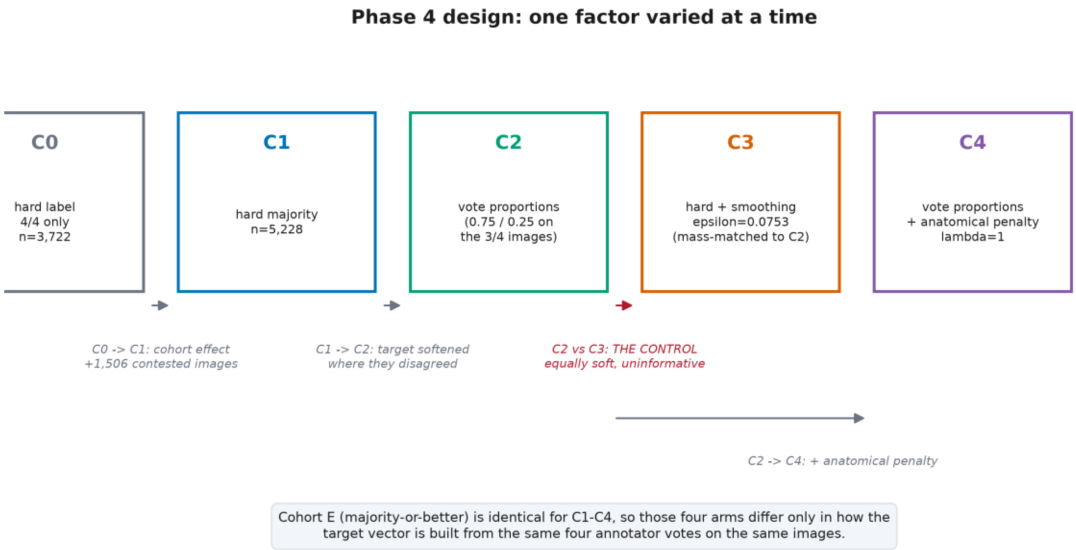


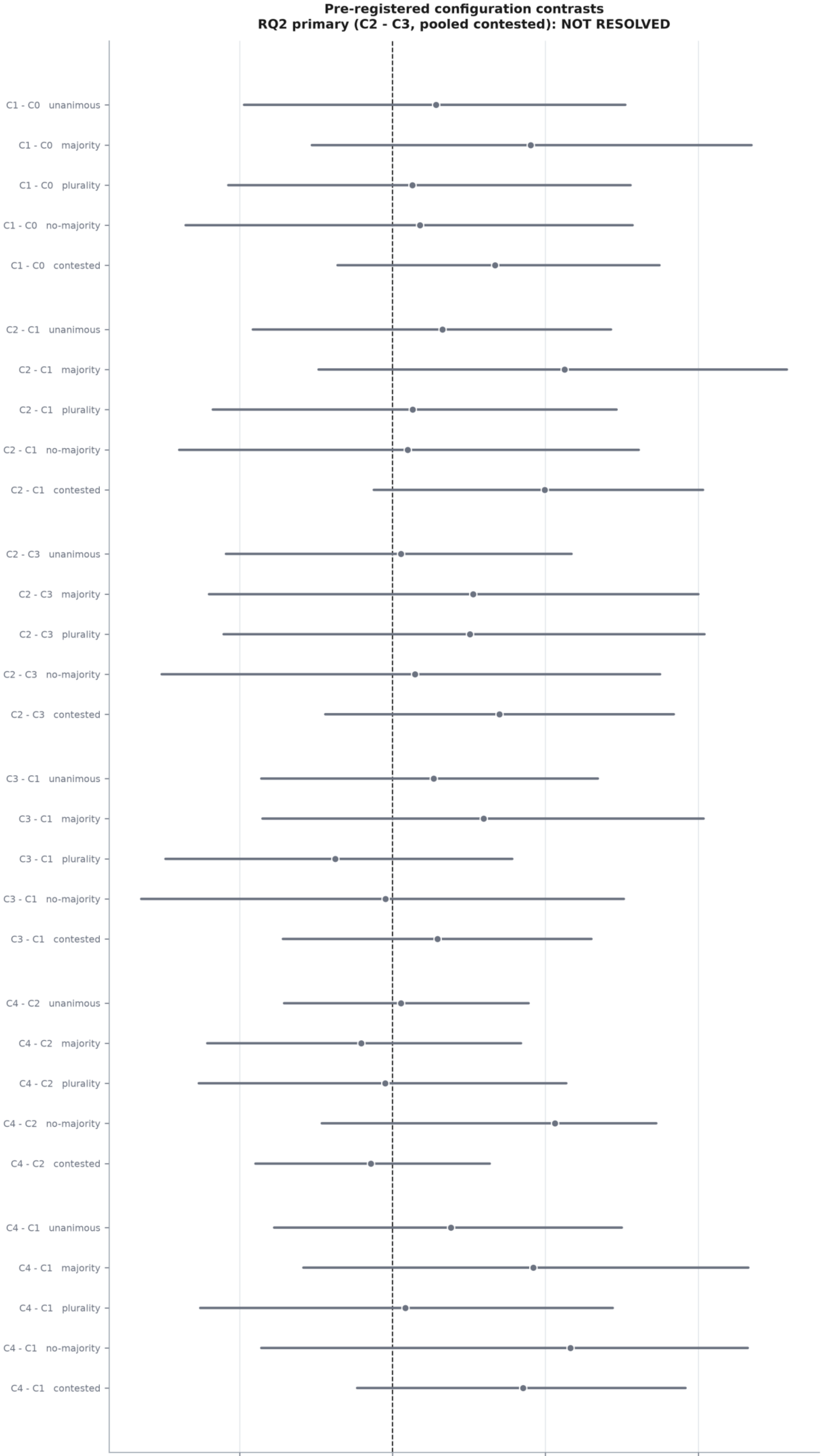
Figure 18. The configuration matrix.

6.2 Deriving the control rather than choosing it

C3 exists so that any benefit of the vote-proportion target can be separated from ordinary regularisation. Its smoothing strength was derived, not set to a convention: $\varepsilon = 0.07529$ matches the probability mass that the soft target displaces from the modal label. Matching by entropy instead would have given 0.021228, a materially weaker control. Mass rather than entropy, because the gradient of the soft-target cross-entropy is $(q - t)$, so the displaced mass is the perturbation.

6.3 Results, and the reversal that is the finding

The pre-registered contrast C2 − C3 on the pooled contested stratum is 1.40 points ([-0.88, 3.68]): not resolved. On calibration the result is stronger and points the other way — the generic control is markedly better calibrated on contested images than the vote-proportion arm, while on unanimous images the order reverses. Uniform smoothing suppresses confidence globally, so it is under-confident where annotators agree and closer to correct where they do not; C2 is trained one-hot on unanimous images and is nearly exact there.



Pre-registered configuration contrasts
RQ2 primary (C2 - C3, pooled contested): NOT RESOLVED
C1 - C0 unanimous
C1 - C0 majority
C1 - C0 plurality
C1 - C0 no-majority
C1 - C0 contested
C2 - C1 unanimous
C2 - C1 majority
C2 - C1 plurality
C2 - C1 no-majority
C2 - C1 contested
C2 - C3 unanimous
C2 - C3 majority
C2 - C3 plurality
C2 - C3 no-majority
C2 - C3 contested
C3 - C1 unanimous
C3 - C1 majority
C3 - C1 plurality
C3 - C1 no-majority
C3 - C1 contested
C4 - C2 unanimous
C4 - C2 majority
C4 - C2 plurality
C4 - C2 no-majority
C4 - C2 contested
C4 - C1 unanimous
C4 - C1 majority
C4 - C1 plurality
C4 - C1 no-majority
C4 - C1 contested

Figure 19. RQ2's pre-registered contrast against the matched control.

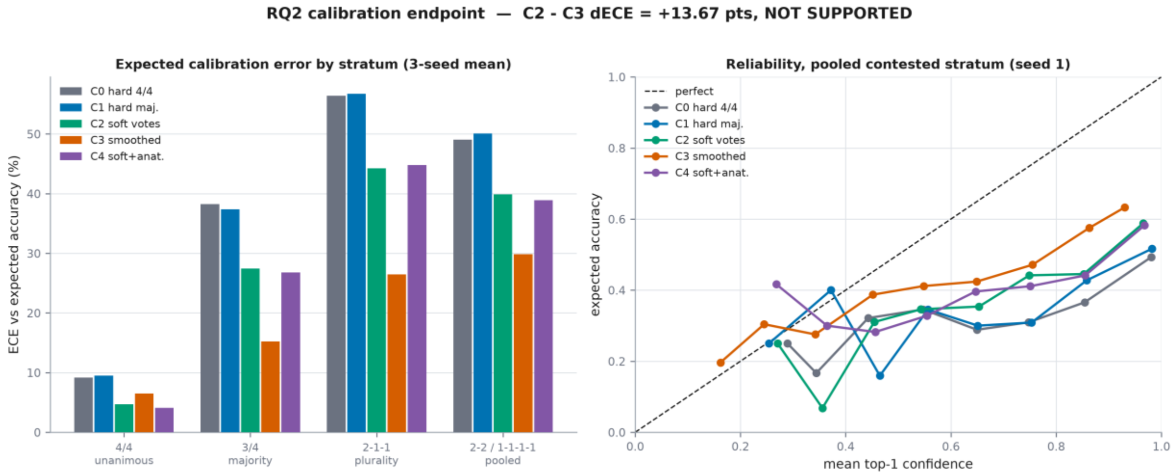


Figure 20. Calibration by configuration and stratum.

> **What the control bought**
> *No configuration achieves acceptable calibration on contested images. That relocates the Chapter 5 finding from 'an artefact of consensus-only training' to a property of the problem, and it is the reason the thesis's recommendation in Chapter 11 is about abstention rather than about target design.*

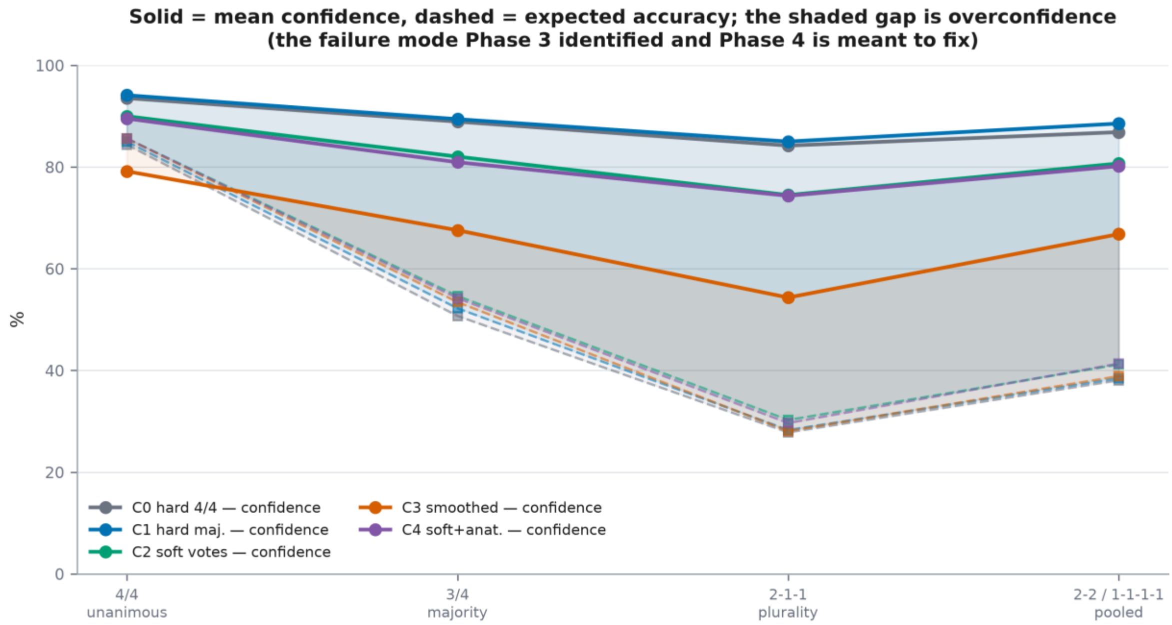


Figure 21. Confidence against accuracy across strata.

6.4 Why RQ2, RQ3 and RQ4 are reported as unresolved

RQ2 is not resolved on accuracy and not supported on calibration. RQ4's anatomy-aware penalty gives -0.00270 ([-0.01068, 0.00469]) at unit λ, and because no λ sweep was run this is evidence about unit weight rather than about anatomy-aware losses in general. RQ3 is addressed in Chapter 8, where it turns out not to be estimable in the form it was posed.

> **A bound that must be declared**
> *Eight of the twelve training runs stopped at the compute-imposed epoch cap rather than by early stopping. The cap applies identically to all four arms, so the target contrasts are unaffected, but the absolute scores are lower bounds and the C1−C0 comparison is not controlled.*

7. External Validation

7.1 The label-space finding, which precedes any transfer number

HyperKvasir [5] and GastroVision [38] were acquired, hashed against the GastroHUN inventory with zero collisions, and mapped by a table frozen before any image was scored. The blueprint's premise did not survive contact with the data: GastroHUN's label space is wall × station, and neither external corpus carries the wall axis or has a class for four of the six stations. A 23-way external validation is not available from these corpora. The phase was therefore reframed — before scoring — into a 2-way anatomical collapse over 3,125 gastric images and an out-of-protocol rejection endpoint over 13,997 images that are not gastric stations at all.

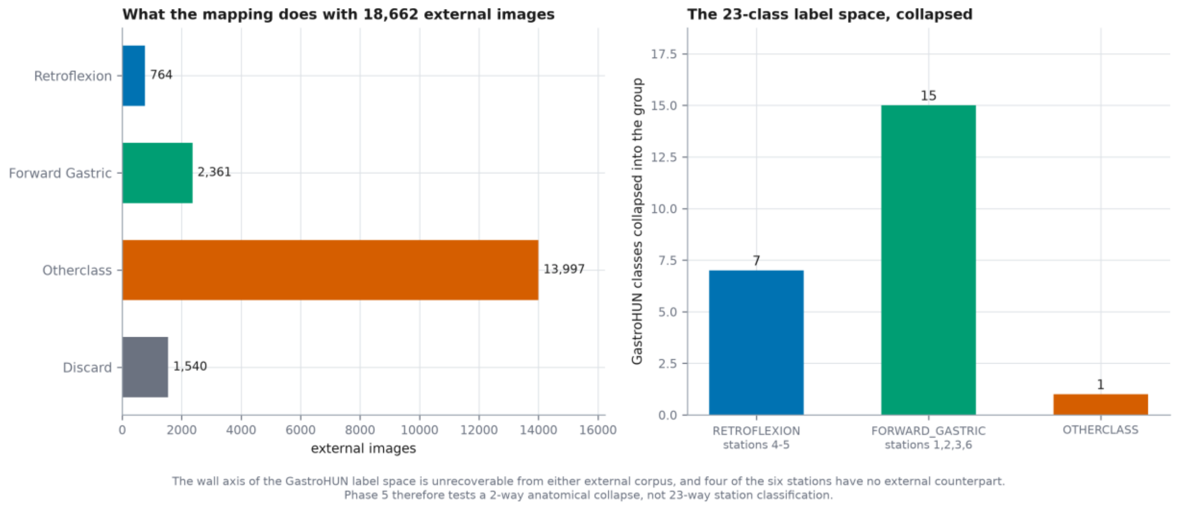


Figure 22. What the mapping destroys.

7.2 Transfer and rejection

Transfer: TRANSFERS. Binary macro F1 78.41 externally against 98.17 internally, a drop of -19.75 points ([-21.29, -18.28]) — about twice the pre-registered expectation. Every arm met the pre-registered precision target, so these are powered verdicts rather than underpowered nulls.

Table 16. Out-of-protocol rejection against a chance floor of 4.35%. The pre-registered hypothesis was that rejection would sit at or below chance; it was falsified favourably, and the soft-target arms reject far better than the hard-label ones.

Arm	Target construction	Out-of-protocol rejection %
C0	C0 hard label, 4/4 cohort	41.9
C1	C1 hard majority label	41.8
C2	C2 vote proportions	63.4
C3	C3 hard + matched smoothing (control)	43.1
C4	C4 vote proportions + anatomical penalty	56.4

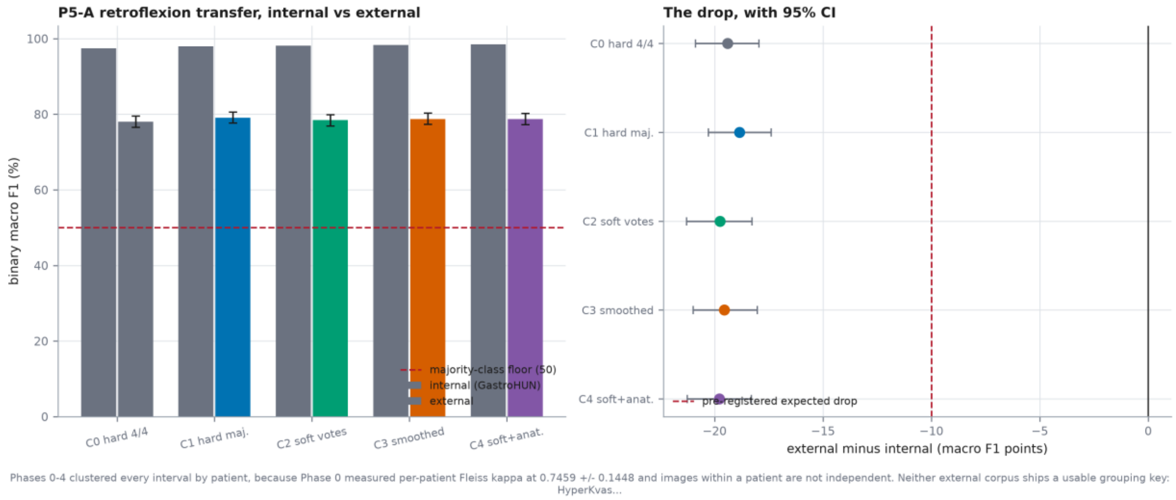


Figure 23. Transfer by arm, internal against external.

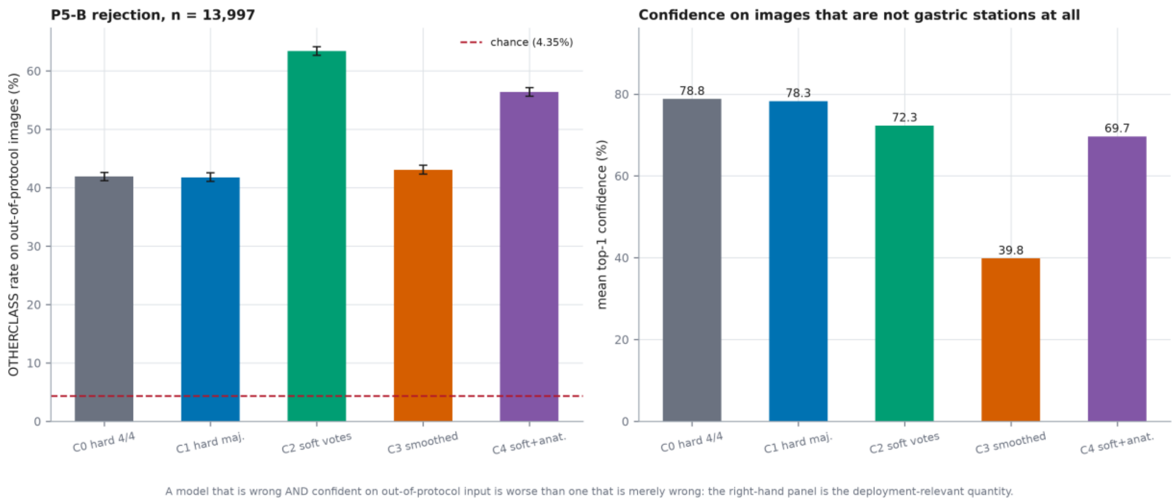


Figure 24. Out-of-protocol rejection by arm.

Why this endpoint required a second centre

This benefit is invisible internally. GastroHUN's test split holds almost no true out-of-protocol images, so the endpoint on which the arms separate most sharply cannot be measured on the corpus the model was trained on. That is the argument for external validation stated as a measurement rather than as a principle.

7.3 Self-training, and what adaptation cost

Table 17. Phase 5B self-training, run only after the clean transfer numbers were frozen and committed — adapting first would have made the external validation circular.

Endpoint	Verdict
P5-A	HURTS
P5-B	ADAPTATION HELPS
P5-C	WORSE CALIBRATED
internal_retention	NO SIGNIFICANT REGRESSION

A declared weakness of the 5B split

the split is image-level because neither corpus publishes a patient or case identifier. Frames from the same procedure can therefore fall on both sides of the split, which makes the ADAPTED arm look better than it would under a patient-level split. Any 5B gain is consequently an UPPER BOUND on what adaptation buys, and the report must say so.

An interval caveat that travels with every external number

All Phase 5 and 5B intervals are image-level, because neither external corpus publishes a case identifier. They are optimistic relative to a correctly clustered interval and must not be compared directly against the patient-clustered intervals used everywhere else in this thesis.

8. Explainability and Error Analysis

8.1 Changing the comparator

Every comparison in Chapters 5 to 7 was model against model. That design ranks target constructions but cannot calibrate the ranking against anything outside it: a model scoring in the twenties looks broken until one knows what a board-certified endoscopist scores on the same images. This chapter introduces two comparators the project had never used — the annotators themselves, and the model's own confidence ordering.

8.2 The human comparator

Each annotator is held out in turn and scored against the other three; the model is scored against the same three, on the same images, under the same patient resample. Annotator a is excluded from their own reference set, because scoring a rater against a panel containing themselves makes one term an identity and inflates the human side.

Table 18. The human comparator with the oracle that bounds it. The singleton rate is the fraction of held-out annotators whose label is shared by none of the other three — a structural handicap fixed by the stratum definition, not by skill.

Stratum	Held-out annotator	Model (C2)	Modal-vote oracle	Headroom recovered	Singleton rate
Unanimous (4/4)	1.0000	0.8514	1.0000	n/a	0%
Majority (3/4)	0.4893	0.5301	0.7423	16%	25%
Plurality (2-1-1)	0.1436	0.2668	0.4492	40%	50%
No majority	0.2506	0.3135	0.5447	21%	10%
Contested (pooled)	0.3902	0.4575	0.6700	24%	29%

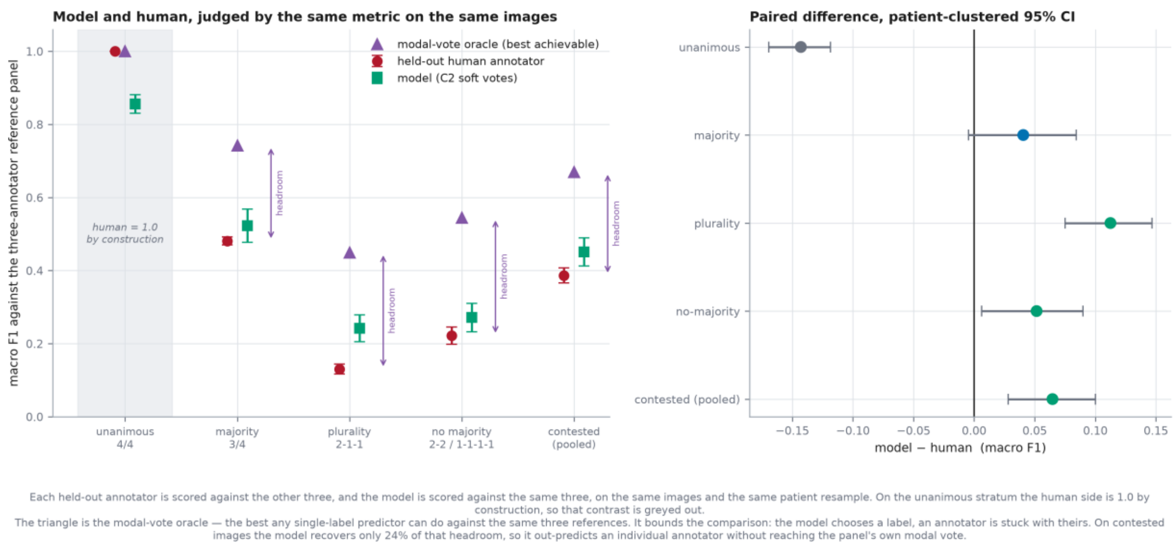


Figure 25. Held-out annotator, model and modal-vote oracle on a common axis.

What may actually be claimed

OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE. The model recovers 24% of the headroom between a held-out annotator (0.3902) and the modal-vote oracle (0.6700). The pre-registered 'ABOVE THE HUMAN PANEL' therefore reflects that predicting a panel is an easier task than being a member of it — the model chooses a label, an annotator is stuck with theirs, and 29% of held-out annotators on this stratum are singletons who cannot score well whatever their skill. It is NOT evidence of superior anatomical judgement, and a substantial model shortfall against the attainable ceiling remains.

The pre-registered rule returned ABOVE THE HUMAN PANEL on the contested strata. That verdict stands as the frozen rule produced it, but it must not be read as 'the model outperforms an expert'. Two asymmetries, found on re-reading the construction and then measured, explain it: the model is optimised to predict this panel's consensus while the annotator is simply being themselves, and — more decisively — the model chooses a label while the annotator is stuck with the one they gave. The modal vote of the same three references bounds any single-label predictor, and the model does not reach that bound on any stratum.

8.3 Confusion geometry, and the settlement of a withdrawn claim

On the pooled contested stratum the model's wall-confusion geometry differs from the human geometry by -0.92 points ([-4.69, 2.78]) and its station geometry by -8.19 points ([-15.10, -2.85]). When the model confuses the circumferential direction of the scope it confuses it the way endoscopists do; when it confuses depth, it travels further along the insertion axis than human disagreements do. That is a specific, nameable deficit rather than a general one.

Settling the withdrawn claim

X3 is settled, and by a stronger argument than the one that raised it. The amendment withdrew the Phase 3 claim because a model INTERVAL was compared against a human POINT estimate. Measuring both sides on the same images shows something more basic: S-unanimous is defined by all four annotators agreeing, so it contains ZERO annotator disagreement events and the human error geometry is UNDEFINED there -- not imprecise, undefined. The Phase 3 comparison therefore set the model's geometry on the unanimous stratum against a human benchmark computed over the whole corpus, which is dominated by contested images. It was a cross-population comparison, and neither the 0.12-point 'match' nor the 7.5-point 'shortfall' was ever a like-for-like quantity.

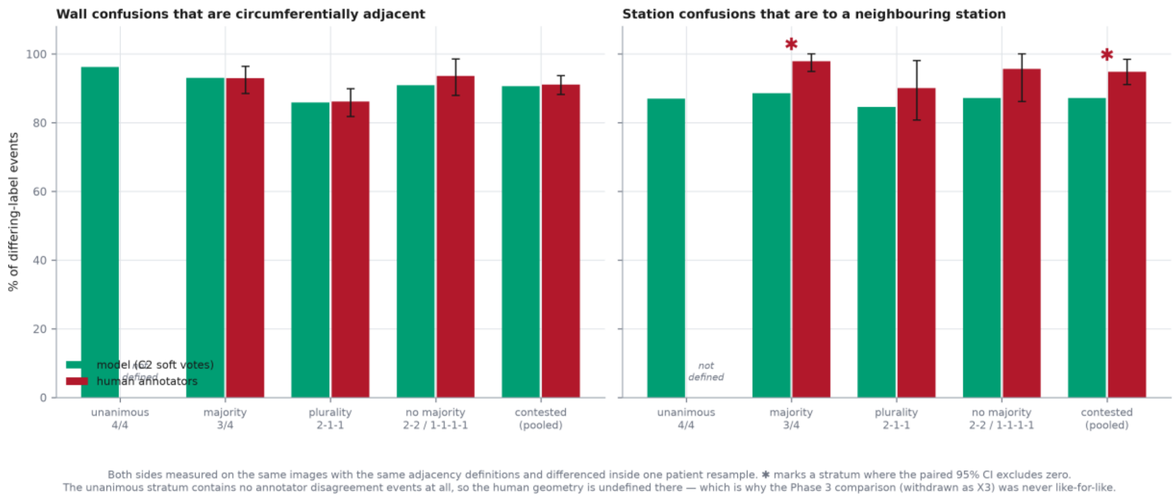


Figure 26. Error geometry, model and human, both with patient-clustered intervals.

8.4 Attribution: a measurement that could not be made

The pre-registered attribution endpoint — the within-stratum correlation [69] between Grad-CAM dispersion and annotator vote entropy — returned NOT ESTIMABLE (vote entropy is constant on this stratum by construction). Vote entropy is a deterministic function of the vote pattern (a 3-1 split is always 0.5623 nats) and the strata are defined by that pattern, so entropy is constant within a stratum and the correlation does not exist. Reporting this as 'no association' would assert a measurement that was never available.

Figure 27. Why the pre-registered attribution signal cannot be computed within a stratum.

Table 19. The secondary attribution endpoint did return a result: attribution destabilises on contested images for every arm, and the three soft-target arms are markedly more spatially consistent across seeds than the two hard-label arms.

Arm	Inter-seed IoU (unanimous)	Verdict
C0	0.486	ATTRIBUTION DESTABILISES ON CONTESTED IMAGES
C1	0.503	ATTRIBUTION DESTABILISES ON CONTESTED IMAGES
C2	0.676	ATTRIBUTION DESTABILISES ON CONTESTED IMAGES
C3	0.678	ATTRIBUTION DESTABILISES ON CONTESTED IMAGES
C4	0.670	ATTRIBUTION DESTABILISES ON CONTESTED IMAGES

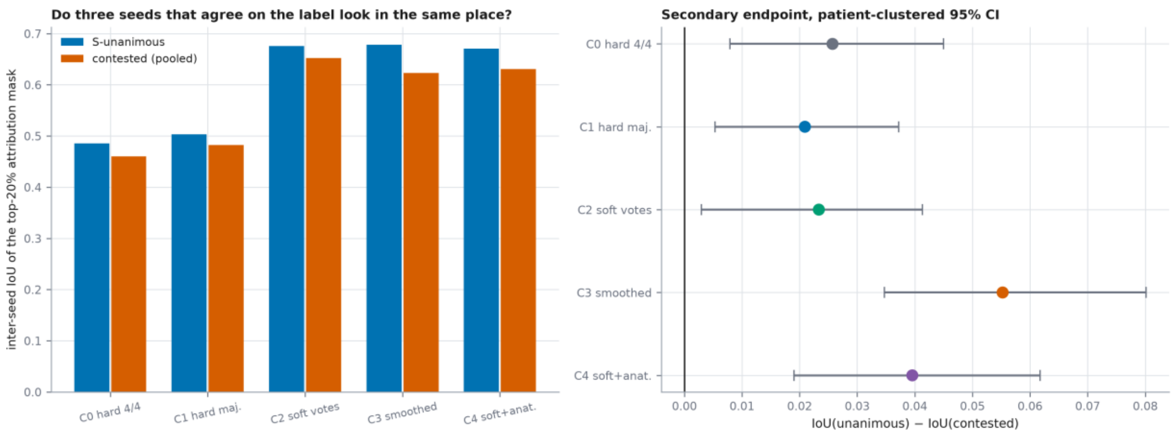


Figure 28. Inter-seed attribution stability.

8.5 Selective prediction

Table 20. Area under the risk–coverage curve, lower is better. Internally the arms are nearly indistinguishable; externally they separate decisively. External intervals are image-level and are not comparable with internal ones.

Arm	Internal AURC	External AURC
C0	0.0977	0.3315
C1	0.0927	0.3382
C2	0.0828	0.1906
C3	0.0822	0.3235
C4	0.0870	0.2319

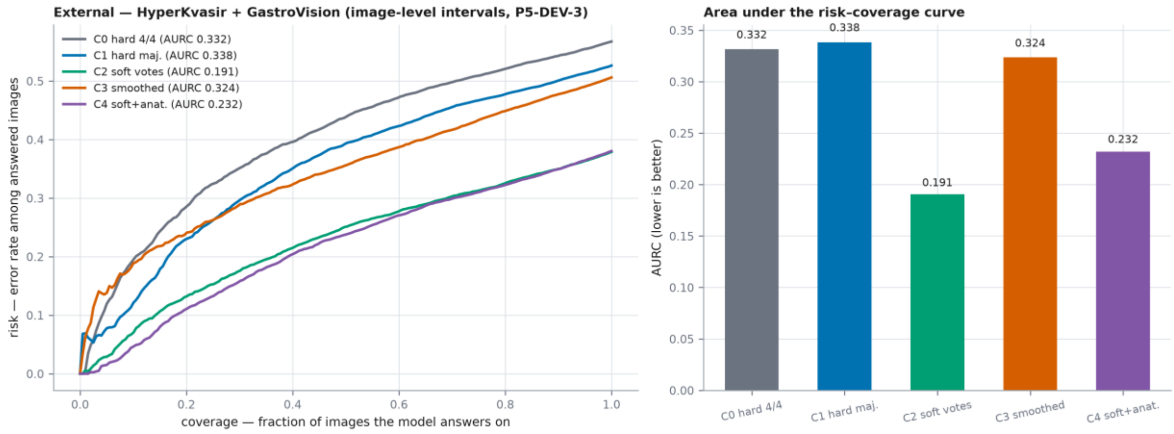
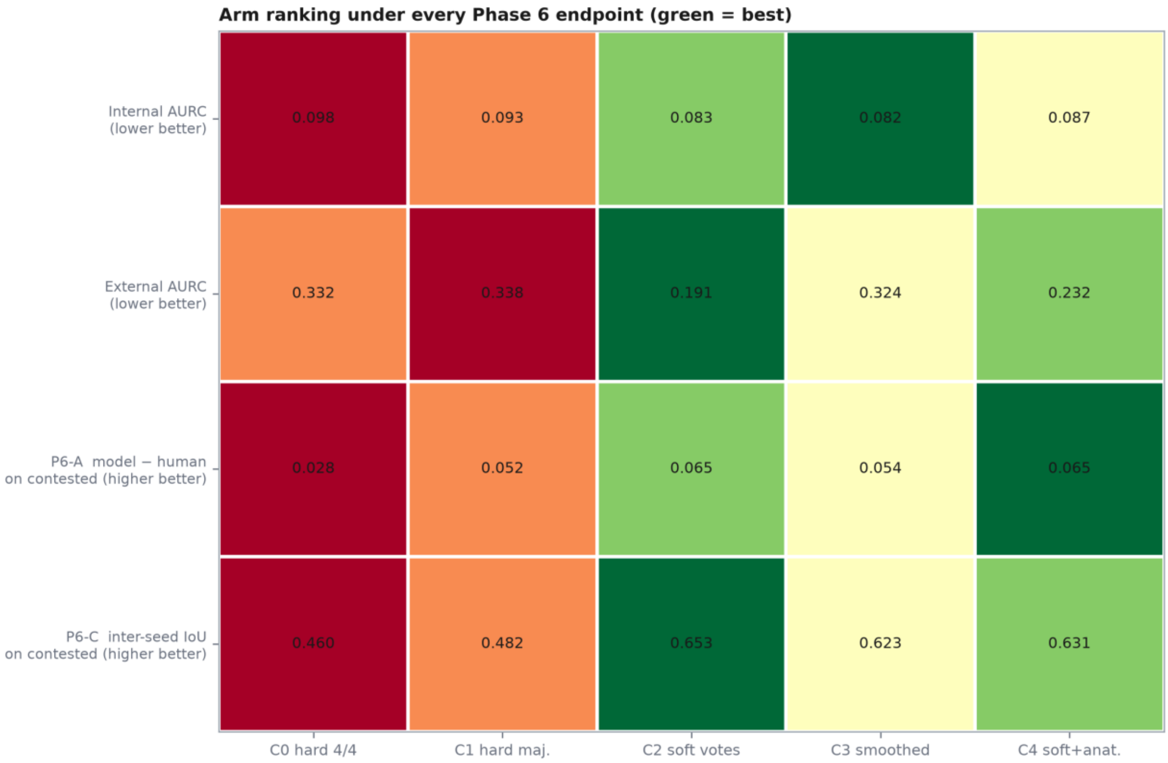


Figure 29. **Risk–coverage on the external panel.**

The ordering agrees with the single-operating-point rejection result of Chapter 7 (CONSISTENT WITH THE PHASE 5 RANKING), so that finding was not an artefact of where the 23-way decision boundary happens to fall.



Cell shading is the rank within the row; the printed value is the measured quantity. Rows disagree, which is the point: no single arm wins everywhere.

Figure 30. **Arm ranking under every endpoint. The rows disagree, which is the point.**

9. The Audit Protocol as an Instrument

9.1 An audit that passes everything it sees is not an audit

Chapter 2's protocol returned PROCEED on the adopted corpus. That result is uninformative unless the protocol can also return the opposite. This chapter scores it against a corpus known to be unsound: the peptic-ulcer dataset this project began with and retired.

9.2 Gate-by-gate comparison

Table 21. The eight gates applied to both corpora, each reduced to a criterion that is meaningful in both modalities and derived from artefacts rather than quoted from report prose.

Gate	Criterion	Adopted corpus	Retired corpus
G1	Provenance, ethics and licence	PASS	FAIL
G2	Physical integrity	PASS	PASS
G3	Duplication and contamination	PASS	PASS
G4	Label architecture	PASS	FAIL
G5	Agreement quantified	PASS	NOT ASSESSABLE
G6	Split integrity	PASS	FAIL
G7	Statistical power	CONDITIONAL	FAIL
G8	Population description	CONDITIONAL	PASS

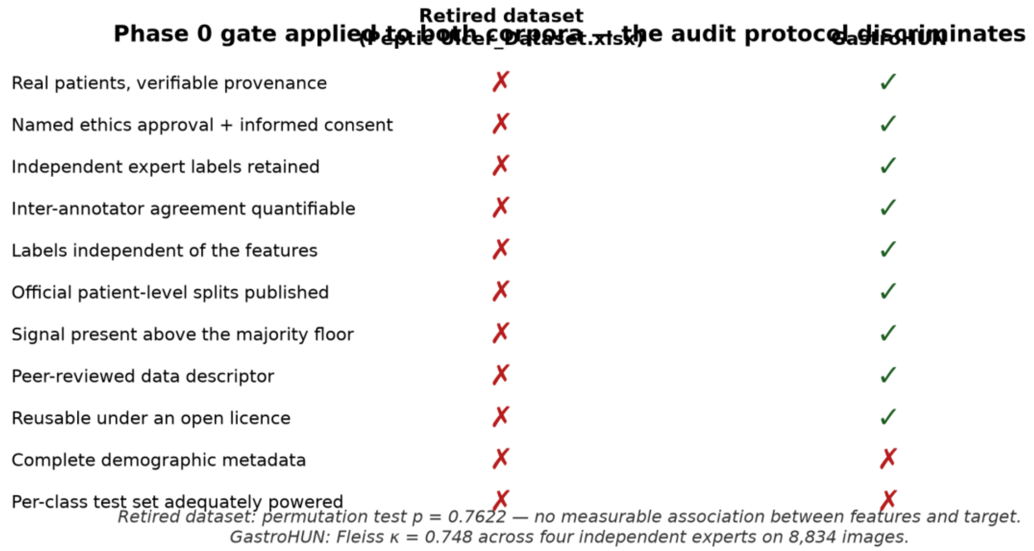


Figure 31. The negative control.

Gate-counting is not a quality score

6 of 8 gates separate the corpora, but the separation is weaker than that count suggests. Only 4 (G1, G4, G6, G7) are outright failures; G5 cannot fire at all on a single-annotator corpus; G2, G3 pass on BOTH; and on G8 the UNSOUND corpus scores HIGHER than the adopted one, because it ships the age and sex that GastroHUN omits. Gate-counting is therefore not a quality score: a report presenting a tally of passes as evidence of soundness would have described the retired corpus favourably on 3 of 8 gates.

9.3 What the protocol missed

Table 22. Only 1 of 4 fatal defects is caught by any gate, and that one only incidentally.

ID	Fatal defect	Caught by	Why not

ID	Fatal defect	Caught by	Why not
D1	Zero predictive signal	none	no gate in G1-G8 measures whether the features carry information about the target. A corpus of pure noise with clean provenance, clean integrity, no duplicates and a full population description passes six of the eight gates.
D2	Circular label construction	G4 (partially)	G4 fails the corpus for having no per-annotator labels, which is a different fault that happens to co-occur. G4 as specified does not test whether the target is derivable from the features, so a corpus with four annotators AND a circular target would pass it.
D5	Consistent with uniform random generation	none	no gate tests whether the data is consistent with having been synthesised. Every marker here -- uniformity, balance, independence, a tiny phrase bank -- is invisible to G1-G8.
D12	No external validity	none	the protocol audits the corpus in isolation and never asks what it could be validated against.

Well-formedness is not viability

every defect that actually killed the retired corpus was found by the Phase 1.5 signal-and-circularity battery, not by the eight-gate protocol. The gates verify that a corpus is well-formed; they do not verify that it is informative. Those are different questions and this thesis conflated them until the negative control was scored.

9.4 Proposed extensions

Table 23. Three extensions, each of which would have caught a fatal defect and each of which costs minutes to run.

ID	Gate	Criterion	Would have caught
G9	Signal existence	feature-target association must exceed a permutation null at a pre-registered level; report the effect size, not only the p-value	D1
G10	Label independence	the target must not be reconstructible from the features by rule. Test by fitting a trivial rule-based predictor and by ablating each feature field in turn	D2
G11	Synthesis markers	test for uniformity of continuous fields, near-exact categorical balance, pairwise independence of clinically coupled fields, and a small closed vocabulary	D5

a negative control that flattered the protocol would have been worth nothing. This one changed the protocol, which is what a negative control is for, and it is the reason RQ5 belongs in the thesis as a chapter rather than as a remark.

10. Synthesis

10.1 The central claim, defended

The claim is that agreement stratification separates a falling reference standard from a falling classifier. Three alternative readings deserve answering.

- 'The model simply fails on hard images.' If that were the whole story, a held-out expert would do well where the model does badly. They do not: on contested images the annotator scores 0.3902 against the model's 0.4575.
- 'The task is simply impossible there, so the model is blameless.' If that were the whole story, the model would sit at the attainable ceiling. It does not: the oracle reaches 0.6700 and the model recovers only 24% of the distance to it.
- 'This is a ConvNeXt artefact.' This was the strongest remaining objection and it has now been tested. 3 of 3 claims replicate on EfficientNet-B0 (4.0 M parameters against ConvNeXt-Tiny's 28 M) — see §10.2a. Three endpoints still rest on one architecture and are named in §10.3.

10.2 What the nulls establish

Three of five research questions returned unresolved or not-estimable verdicts. Reported carelessly that is a weak thesis; reported precisely it is the result. The nulls are informative because each was a pre-registered test against a matched control, with a precision target met, rather than an underpowered search that happened to find nothing. That no target construction repairs calibration is a stronger statement than that one of them did, because it moves the problem from the training objective to the problem itself — and it is what licenses the recommendation to deploy abstention rather than a better loss.

Multiplicity

Of the 5 primary endpoints, 3 are sampling statistics with intervals. 1 of those exclude zero unadjusted. Adjustment can only widen an interval, so the 2 endpoints that already contain zero are unaffected -- RQ2 and RQ4 were reported as unresolved and remain so. 1 of 1 survive Holm at family-wise alpha 0.05. The thesis's confirmatory claims are therefore unchanged by multiplicity, which is the useful thing to be able to state and could not be stated before this analysis existed.

10.2a Backbone generalisation

To test whether these conclusions are properties of ConvNeXt, the target-construction contrast was re-run on EfficientNet-B0 (4.0 M parameters against ConvNeXt-Tiny's 28 M) [74]. The training script imports the Phase 4 code and rebinds only the model constructor, so cohort, cache, normalisation, augmentation, schedule, loss, early stopping, epoch cap and seeds are identical and any difference is architectural. Fine-tuning depth was matched in parameter terms rather than module count (92.3% of feature parameters unfrozen against ConvNeXt's 94.5%). Every metric was computed by the same function that produced the ConvNeXt number.

Table 24. Backbone replication. 3 of 3 claims hold on an architecture with one seventh the parameters and a different inductive bias.

	Claim	ConvNeXt-Tiny (28 M)	EfficientNet-B0 (4 M)	Replicates
R1	RQ2 accuracy contrast C2−C3 on contested images	NOT RESOLVED (1.40 [-0.88, 3.68])	NOT RESOLVED (0.67 [-0.55, 1.90])	yes
R2	Calibration reversal: C3 better on contested, C2 on unanimous	present = True	present = True	yes
R3	Model between held-out annotator and modal-vote oracle	24% of headroom; exceeds oracle = False	20% of headroom; exceeds oracle = False	yes

The calibration reversal is the thesis's durable finding and it reproduces in both directions. On EfficientNet-B0 the matched control C3 is better calibrated than the vote-proportion arm C2 on contested images (21.8% against 29.4%), and the order reverses on unanimous images (3.5% against 5.3%) — the same pattern, at the same strata, in a network with one seventh the parameters.

The limit of this replication
What this does NOT cover: confusion geometry (Chapter 7 §7.3); attribution stability (Chapter 7 §7.4); external transfer, rejection and selective prediction (Chapters 6, 7 §7.5). Those remain single-architecture results and the threat table below is updated accordingly rather than declaring the objection closed outright.

10.3 Threats to validity, ranked

Table 25. Threats to validity, ranked by how much they could change a conclusion.

Threat	Severity	Status
Single backbone for the geometry, attribution and external endpoints	Medium	Reduced but not eliminated: the accuracy null, the calibration reversal and the human-comparator position replicate on EfficientNet-B0 (§10.2a); the three endpoints named there do not yet have a second backbone.
Compute-bound epoch cap on 8/12 runs	Medium	Applies identically across arms, so contrasts hold, but absolute scores are lower bounds.
Contested strata are small (n = 342, 127, 81)	Medium	Per-stratum intervals are wide; the pooled stratum carries the endpoints.
External intervals are image-level	Medium	No external corpus publishes a case key; declared wherever an external number appears.
Four annotators bound the human comparator	Medium	The held-out construction leaves a reference panel of three.
Strata are defined by the same agreement that scores the human	Medium	The model-minus-human contrast is unaffected (identical rows), but the human curve alone is not a free-standing estimate of expert accuracy.
No age or sex in the corpus	Low for the claims made	No demographic or fairness claim is made anywhere.
Confirmatory family defined retrospectively	Low	Each RQ's primary endpoint was pre-registered as that RQ's test; only the grouping is new.

10.4 What could be deployed, and under what conditions

Nothing in this thesis supports deploying an autonomous landmark classifier. What it does support is a completeness-checking assistant that abstains. The external risk–coverage analysis gives the operating characteristic: the C2 arm reaches an AURC of 0.1906 against 0.3235 for the internally-best arm, and its out-of-protocol rejection is the only endpoint in the project that separates configurations under domain shift. A deployment would have to select on that endpoint, monitor it at the new centre, and route declined images to a human rather than scoring them.

11. Conclusions and Future Work

11.1 Answers to the research questions

Table 26. Answers to the five research questions. Every verdict was selected by a rule frozen before the analysis ran.

RQ	Question	Answer
RQ1	Does performance vary across strata of expert agreement?	Yes, and the ceiling moves with it. Ceiling-normalised gap 17.98 pts [12.49, 24.37].
RQ2	Do soft targets from all four votes beat hard consensus labels?	NOT RESOLVED on accuracy (1.40 pts [-0.88, 3.68]); NOT SUPPORTED on calibration — the control wins on contested images.
RQ3	Does predictive uncertainty track human disagreement?	Supported for no configuration internally, and NOT ESTIMABLE in the within-stratum form it was posed, for a structural reason (Chapter 8).
RQ4	Does an anatomy-aware loss help on contested images?	NOT RESOLVED at unit λ (-0.00270 [-0.01068, 0.00469]). No sweep was run, so this is evidence about unit weight only.
RQ5	Does the audit protocol discriminate a sound corpus from an unsound one?	PARTIALLY SUPPORTED — 6/8 gates separate the corpora but only 1/4 fatal defects is caught by any gate.

11.2 Contributions restated

Against the reporting gaps identified in Chapter 4, this thesis contributes an agreement-stratified protocol that reports the attainable ceiling alongside the score; a human comparator and oracle that make the model's residual shortfall measurable rather than assumed; evidence that the calibration failure is a property of the problem rather than of the training objective; a demonstration that selective prediction separates configurations that no internal endpoint can; and a negative-control evaluation that improved the audit protocol it tested.

11.3 Future work, costed

Table 27. Future work, costed. Items marked not possible are limits of the available data, not of effort.

Work	Cost	What it would buy
Second backbone across the geometry, attribution and external endpoints	~8 GPU hours	The target contrast, calibration reversal and human-comparator position already replicate (§10.2a); these three do not yet.
λ sweep for the anatomy-aware loss	~15 GPU hours	Converts RQ4's unit-λ null into an answer.
Raise the epoch cap on all twelve Phase 4 runs	~22 GPU hours	Removes the lower-bound caveat on absolute scores.
Attribution-method sweep	~45 minutes	Deliberately not run: it would invite selecting whichever method correlated best.
Patient-clustered external intervals	not possible	Neither external corpus publishes a case identifier.
Human comparator on external corpora	not possible	Neither external corpus publishes per-annotator labels.
Prospective evaluation of the abstaining assistant	a clinical study	The only way to test the deployment claim of §10.4.

Appendix A. Pre-registration records

Table 28. Each pre-registration script refuses to overwrite an existing file.

Phase	Artefact	What it fixed
Phase 2	reports/phase2_prereg.json	target, seeds, bootstrap, diagnostic order
Phase 4	reports/phase4_prereg.json	ε derived, λ fixed, verdict rules for RQ2–RQ4
Phase 5	reports/phase5_prereg.json	collapse, endpoints, precision target
Phase 6	reports/phase6_prereg.json	four endpoints, CAM layer, top-q, verdict rules

Appendix B. Declared deviations and amendments

Table 29. Every deviation was declared before or at the moment it was taken, and every amendment was forced by a gate or by a numerical fact rather than adopted silently.

ID	Deviation or amendment
P4-DEV-1	MC dropout → MC stochastic depth (ConvNeXt has no Dropout/BatchNorm)
P4-DEV-2	5-member ensemble → 3 members, on budget
P4-DEV-3	λ fixed at 1.0, no sweep
P4-DEV-4	3 dataloader workers → 2, after a reproducible CUDA host-allocation failure
P5-DEV-3	External intervals image-level; no case key exists
X1–X4	Four Phase 3 claims withdrawn on re-examination (Chapter 5 §5.4)
P6-DEV-1	Grad-CAM on all images, not only contested ones
P6-DEV-2	Human comparator and selective prediction added to Phase 6
P6-DEV-3	Grad-CAM only; no attribution-method sweep
P6-AMD-1	Human comparator degenerate on a unanimity-defined stratum
P6-AMD-2	Human error geometry undefined on S-unanimous
P6-AMD-3	CAM targets the committed prediction, not the live argmax
P6-AMD-4	Vote entropy constant within a stratum; RQ3 endpoint not estimable
P6-AMD-5	Exposure and choice asymmetries in the human comparator

Appendix C. Multiplicity

Table 30. one primary endpoint per research question, listed below. These and only these carry a family-wise error guarantee. EXPLORATORY. This includes every per-class metric, every per-seed value, every stratum other than the primary one for its RQ, all Phase 5 external endpoints, all Phase 6 secondary endpoints, and all sensitivity analyses. Exploratory results may be reported and discussed but may not be described as confirmed.

RQ	Primary endpoint	Under multiplicity
RQ1	ceiling-normalised macro F1 gap, S-unanimous minus S-majority (Phase 3B)	excludes 0; Holm survives = True
RQ2	annotator-marginalized macro F1, C2 minus C3, on the pooled contested stratum (Phase 4)	contains 0
RQ3	within-stratum Spearman rho between model uncertainty and annotator vote entropy (Phase 4 predictive; Phase 6 spatial)	not estimable
RQ4	expected anatomical error distance, C4 minus C2, at lambda = 1, on the pooled contested stratum (Phase 4)	contains 0
RQ5	number of modality-independent gates separating the sound from the unsound corpus, and the number of fatal defects caught by any gate	no interval

Appendix D. Figure provenance

Table 31. figures are SELECTED and RENUMBERED, never redrawn. Redrawing would create a second code path capable of disagreeing with the phase figure it duplicates. Each thesis figure traces to the phase script that produced it, recorded below.

Figure	Chapter	Phase figure	Drawn by
T01	Ch 1	F19_conceptual_framework.png	src/report/figures_v2.py
T02	Ch 1	F01_thesis_workflow.png	src/report/figures_v2.py
T03	Ch 2	F03_sss_taxonomy.png	src/report/figures_v2.py
T04	Ch 2	F05_kappa_matrix.png	src/report/figures_v2.py
T05	Ch 2	F06_agreement_cascade.png	src/report/figures_v2.py
T06	Ch 2	F07_disagreement_decomposition.png	src/report/figures_v2.py
T07	Ch 2	F09_otherclass.png	src/report/figures_v2.py
T08	Ch 2	F15b_dup_calibration.png	src/report/figures_v2.py
T09	Ch 2	F13_test_power.png	src/report/figures_v2.py
T10	Ch 3	F17_prisma.png	src/report/figures_v2.py
T11	Ch 3	F18_literature.png	src/report/figures_v2.py
T12	Ch 4	P2_F10_verdict.png	src/report/figures_phase2.py
T13	Ch 4	P3_F21_stratified_curve.png	src/report/figures_phase3.py (and figures_phase3b.py)
T14	Ch 4	P3_F25_ceiling_normalised_curve.png	src/report/figures_phase3.py (and figures_phase3b.py)
T15	Ch 4	P3_F26_pairwise_gap_forest.png	src/report/figures_phase3.py (and figures_phase3b.py)
T16	Ch 4	P3_F27_calibration_by_stratum.png	src/report/figures_phase3.py (and figures_phase3b.py)
T17	Ch 4	P3_F30_confound_controls.png	src/report/figures_phase3.py (and figures_phase3b.py)
T18	Ch 5	P4_F25_design.png	src/report/figures_phase4.py
T19	Ch 5	P4_F27_contrast_forest.png	src/report/figures_phase4.py
T20	Ch 5	P4_F28_calibration.png	src/report/figures_phase4.py
T21	Ch 5	P4_F29_overconfidence.png	src/report/figures_phase4.py
T22	Ch 6	P5_F33_label_space.png	src/report/figures_phase5.py
T23	Ch 6	P5_F35_transfer.png	src/report/figures_phase5.py
T24	Ch 6	P5_F36_rejection.png	src/report/figures_phase5.py
T25	Ch 7	P6_F38_human_comparator.png	src/report/figures_phase6.py
T26	Ch 7	P6_F39_confusion_geometry.png	src/report/figures_phase6.py
T27	Ch 7	P6_F40_dispersion_vs_entropy.png	src/report/figures_phase6.py
T28	Ch 7	P6_F41_attribution_stability.png	src/report/figures_phase6.py
T29	Ch 7	P6_F44_risk_coverage_external.png	src/report/figures_phase6.py
T30	Ch 7	P6_F45_synthesis.png	src/report/figures_phase6.py
T31	Ch 8	F16_negative_control.png	src/report/figures_v2.py

Appendix E. Reproducibility index

Every number in this thesis was resolved from a JSON artefact by src/report/phase7_register.py, which read 175 values from 24 artefacts. Running the phase pipelines in order regenerates every artefact, and rebuilding this document from them regenerates every sentence that contains a number.

Table 32. Script manifest.

Phase	Scripts
Phase 0–1	src/data/gastrohun_*.py, src/literature/*_v2.py
Phase 2	src/models/phase2_*.py
Phase 3	src/models/phase3_*.py, phase3b_*.py
Phase 4	src/models/phase4_*.py
Phase 5	src/models/phase5_*.py, phase5b_*.py
Phase 6	src/models/phase6_*.py
Phase 7	src/models/phase7_*.py, src/report/phase7_register.py, figures_thesis.py, build_thesis_docx.py

Appendix F. Citation provenance

The reference list holds 86 entries and none of them was typed into this document. The builder calls cite(key); src/report/bibliography.py resolves the key to a number against reports/phase8_bibliography.json, which is generated from literature_v2/extraction_table.csv. In-text markers and the reference list therefore cannot disagree, and renumbering is automatic.

Table 33. Citation provenance. The two sets are numbered into one list but counted apart, so that the PRISMA total of 82 included studies is never inflated by a reference that did not come through screening — the corpus descriptor, the reporting guideline and the two method papers were never candidates for review inclusion.

Set	In list	Cited in text	Source artefact
Review set (PRISMA-included)	82	82	literature_v2/extraction_table.csv
Additional (corpus, guideline, methods)	4	4	literature_v2/additional_references.json

References

Entries 1–86, ordered by first author surname. Entries drawn from the PRISMA review are marked in Appendix F; the remainder are the corpus descriptor, the reporting guideline this review followed, and two method papers cited for the techniques they define.

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